Housing Dereliction in a Superstar City: Speculative Neglect and Its Disamenity Effects*

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Abstract

Why do derelict houses persist in urban areas despite strong demand? What are their effects on neighbourhoods? This paper proposes a simple mechanism — speculation over future real estate value increases — to explain dereliction amid prosperity and to quantify its disamenity effects. First, we develop a dynamic discrete choice model showing that expectations of future urban regeneration can incentivise property owners to delay redevelopment, thereby prolonging dereliction. Using parcel-level GIS data from central Tokyo, we find that properties located in areas designated for future regeneration are 7–16 percentage points more likely to be derelict. Second, we estimate the effect of derelict houses on nearby rents using future regeneration as an instrument — affecting current rents only through dereliction. Our 2SLS results show that one additional derelict house within 80 metres reduces rents by 1.6% on average. The estimated effect is magnified to 3.6–6.5% in areas with low accessibility to public safety services, suggesting that renters' concerns about fire and crime risks amplify the disamenity effects. Our findings indicate that rational forward-looking behaviour can reduce effective housing supply and generate negative neighbourhood externalities, pointing to unintended consequences of place-based urban policies.

Keywords: Empty dwellings, housing vacancy, supply constraint, waiting behaviour, discrete choice model, hedonic analysis, instrumental variable, urban disamenity

JEL classification: D62, R11, R21, R31, R52

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1 Introduction

Despite chronic housing shortages in superstar cities (Gyourko, Mayer and Sinai, 2013; Hilber and Mense, 2025), a paradoxical trend has emerged — a significant number of properties in these cities remain empty, inactive, or fall into dereliction (Wilson and Barton, 2023; Segú, 2020; Kanayama and Sadayuki, 2021). For instance, long-term empty homes accounted for 3.36% of the total housing stock in the City of London, 1.84% in New York City, and 2.55% in Tokyo, respectively, in 2023.[1] This puzzling phenomenon represents a waste of valuable real estate, which could otherwise accommodate residents and even stabilise housing prices, in areas where housing demand is most acute. While long-term vacancies in declining areas have been well-examined (for example, Molloy, 2016), causes of such vacancies in high-demand cities are still unexplored.

In addition to this resource waste, the persistence of unattended housing in high-demand areas creates another form of harm — disamenity in the surrounding neighbourhood. Inadequate property maintenance poses a threat to neighbourhood safety by reducing "eyes on the street", increasing risks of fires and criminal activities (Ellen, Lacoe and Sharygin, 2013; Cui and Walsh, 2015; Chang and Jacobson, 2017).[2] Moreover, empty properties typically deteriorate over time, creating physical risks of building collapse, vermin, and overgrown vegetation as well as simply being an eyesore. Therefore, housing vacancy and subsequent dereliction could be associated with current and potential residents' concerns about neighbourhood quality (Sadayuki, Kanayama and Arimura, 2020; Suzuki, Hino and Muto, 2022; Paredes and Skidmore, 2017; Whitaker and Fitzpatrick, 2013). Quantifying such disamenity effects is crucial for understanding the extent to which public interventions could be justified. However, causal inference in this context has been methodologically challenging due to endogenous derelict house locations.

This study proposes a simple mechanism — speculation over future real estate value increases — that explains the paradox of housing dereliction in high-demand urban markets and develops an empirical strategy to identify the disamenity effects of derelict properties. First, given that strong demand drives urban regeneration projects (Carozzi, 2020), we shed light on future expectations of such projects as a prosperity-driven cause of today's housing dereliction. We develop a dynamic discrete choice model showing that expectations of future urban regeneration can incentivise property owners to delay redevelopment, thereby prolonging dereliction. Economic intuition is that, when a greater revenue increase is expected in the future, the net benefit of waiting becomes greater, resulting in higher probability of today's left behind derelict houses. Using variation induced by urban regeneration plans in central Tokyo, which provides an ideal setting for our analysis (see Section 3), we find that properties located in designated regeneration areas are 7–16 percentage points more likely to

[1] Data sources: Live tables on dwelling stock (including vacants) by the UK Ministry of Housing, Communities and Local Government for London; American Housing Survey by the US Department of Housing and Urban Development for New York; the Housing and Land Survey by the Statistics Bureau of Japan for Tokyo. Long-term empty dwellings are defined as properties empty for more than six months, except for Tokyo where duration data are unavailable and properties neither for sale nor rent are used as a proxy.

[2] For instance, there are cases where a murderer hides a body on the premises of an empty house, or a fraud group uses an empty house as its hideout. To address these issues, governments have implemented revitalisation policies (Spader, Schuetz and Cortes, 2016), demolition programmes (Paredes and Skidmore, 2017), and registration ordinances (Biswas, Cunningham, Gerardi and Sexton, 2021).

be derelict. Second, we estimate the effect of derelict houses on nearby rents using future regeneration plans as an instrument — affecting dereliction but not current rents through other channels.[3] Our 2SLS results show that one additional derelict house within 80 metres reduces rents by 1.6% on average. The effect is magnified to up to 6.5% in areas with low accessibility to public safety services, suggesting that renters' concerns about fire and crime risks amplify the disamenity effects of derelict properties.[4]

Our findings suggest that rational forward-looking behaviour can reduce effective housing supply and generate negative neighbourhood externalities, highlighting an unintended consequence of place-based urban regeneration policies. Many economics studies show that place-based policies, including transport improvements, housing development, and treatment of lagging regions, could improve welfare under certain conditions (Tsivanidis, 2026; Koster and van Ommeren, 2019; Kline and Moretti, 2014). In contrast, our findings suggest that the announcement of a place-based policy may cause unintended consequences during the pre-implementation period.

Our study relates to the literature on housing vacancy in prosperous areas (i.e., areas with strong housing demand due to population growth, economic vitality, and proximity to central business districts). While much of the literature on housing vacancy has developed focusing on blighted areas (O'Flaherty, 1990; Scafidi, Schill, Wachter and Culhane, 1998; White, 1986; Whitaker and Fitzpatrick, 2013; Paredes and Skidmore, 2017), there have been a limited number of studies on housing vacancy in prosperous areas, particularly on inactive vacancy and dereliction (Segú, 2020; Kanayama and Sadayuki, 2021; Moszkowski and Stackman, 2023). While these studies provide insights into owners' inherent preferences as a cause of inactive vacancy and dereliction despite prosperity, the role of external shocks, such as future policies, that may lead to property neglect despite strong demand has yet to be explored. By examining the direct, first-order effects of urban regeneration policies on today's waiting behaviour, we show that an unintentional consequence of urban policies — the paradox of housing dereliction amid prosperity — can arise.

This study also relates to the literature on optimal timing of irreversible development. The rich literature on real options in real estate development provides valuable insights in this regard. For instance, uncertainty is associated with the delayed timing of irreversible development (Capozza and Li, 1994; Cunningham, 2006; Bulan, Mayer and Somerville, 2009; McMillen and O'Sullivan, 2013; Lin, 2024). Moreover, even with certainty, vacant land development is optimally delayed if cash flows are growing and capital intensity is variable (Capozza and Li, 2002; Lange and Teulings, 2024). While the option value research is highly significant and well-established, it focuses on aggregate price expectations with a limited interest in more granular sources of waiting behaviour.

[3] We show that endogenous locations of future regeneration plans are unlikely, and we can also reject other potential exclusion restriction violations (see Section 5).

[4] These results complement existing studies on urban fire and crime, such as Kim and Yoon (2025), Kallberg and Shimizu (2026), Ellen et al. (2013), and Gibbons (2004). They also have commonalities with a statement made by Don Foster, who was then the UK's Parliamentary Under-Secretary of State for Communities and Local Government, at the National Empty Homes conference 2012: "The empty edifice becomes an eyesore. The eyesore becomes a magnet for rats, vandals, drug dealing, squatters and prostitution. Neighbours are inconvenienced. Communities are damaged. Council and local emergency services stretched to breaking point. The statistics prove the point. Research by The Royal Institution of Chartered Surveyors suggests properties adjoining poorly maintained empty properties can drop by almost 20% in price." This statement is available at https://www.gov.uk/government/speeches/national-empty-homes-conference-2012 (retrieved on 7th April 2026).

Our study complements this literature in two respects. First, while the real options literature has predominantly focused on vacant land, we examine whether the same mechanism — future price expectations delaying redevelopment — operates in the context of vacant housing, which has received limited attention despite generating stronger negative externalities than undeveloped land (Paredes and Skidmore, 2017). Second, by exploiting within-city variation in future expectations arising from a spatially targeted urban policy, we are able to isolate price expectation effects from city-level aggregate trends, going beyond the coarser geographic units typically employed in option value studies.[5]

Finally, this study relates to the literature on the disamenity effects of derelict property. While quantifying these effects is crucial when considering public interventions or understanding unintended effects of policy-induced dereliction, it has been methodologically challenging due to the endogeneity of derelict house locations and the lack of sufficient panel data (Sadayuki et al., 2020; Suzuki et al., 2022; Paredes and Skidmore, 2017; Whitaker and Fitzpatrick, 2013). This is in stark contrast with the foreclosure context (Harding, Rosenblatt and Yao, 2009; Campbell, Giglio and Pathak, 2011; Gerardi, Rosenblatt, Willen and Yao, 2015), where detailed information on repeat sales and foreclosures have allowed researchers to make causal inference. By using a novel instrumental variable in the rental market, we offer the first causal evidence of the social welfare losses from derelict house disamenities. Our identification relies on the fact that future regeneration plans affect housing vacancy in the area but not current rents directly, whereas this exclusion restriction would be violated for sales prices due to capitalisation of future rental streams. Importantly, this approach enables credible causal inference with cross-sectional data, even addressing a key limitation in the foreclosure literature that has largely relied on panel data for identification.[6]

The remainder of this paper is organised as follows. Section 2 develops our theoretical framework. Section 3 introduces the study area and the institutional context, and Section 4 describes the data. Our empirical methodology is outlined in Section 5, followed by a discussion of the empirical findings in Section 6. Finally, Section 7 concludes.

2 Theoretical Model

2.1 Theoretical Framework

We introduce a theoretical framework to guide our empirical analysis. The fundamental question is "why do housing owners in a superstar city leave their houses derelict despite strong housing demand?" To illustrate the mechanism behind this phenomenon specifically to a superstar city, we highlight prosperity-driven factors that have been overlooked in the declining city literature. In a superstar-city setting — where prosperity attracts further economic development — housing owners' decisions are largely influenced not only by today's but also by future economic conditions.

[5]For comparison, Moszkowski and Stackman (2023) divide the south Manhattan into 8 community districts, where each area is approximately 4 to 5 km^2, and Bulan et al. (2009) divide Vancouver into seven areas, where each area is approximately the same size as Toshima City (13km^2).

[6]Compelling instruments are rare in the well-studied US foreclosure context (Harding et al., 2009).

As a source of this dynamic setting, we highlight urban regeneration projects springing up citywide. Housing owners weigh the benefit from immediate redevelopment against the prospect that a structure built today would itself be torn down once urban regeneration raises the permissible building size or forces removal. Economic intuition is that, when an increase in permissible size or forced removal is expected, the return to rebuilding today is eroded, so the net benefit of waiting becomes greater, resulting in a higher probability that houses are left derelict today. In this section, we attempt to show that this intuition holds mathematically and is robust to the types of expectations.

We develop a simple two-period model of teardowns and redevelopment,[7] in which absentee owners of derelict houses decide whether and when to rebuild and rent out their properties. We assume that derelict houses cannot be rented out unless they are rebuilt, as they are not safe to live in. In the first period, an owner i decides — taking into account future expectations — whether and when to rebuild their house. If the owner tears down the house and builds a new house, they can profit from their newly built house until the end of the second period.[8] Specifically, they obtain rent proportional to building size, $r_t k_t$, but incur rebuilding cost of ck_t, where r is unit rent ($r > 0$), k is building height ($k > 0$), c is unit cost of rebuilding ($c > 0$), and $t \in \{1, 2\}$ represents the time period. Let x denote the owner's decision in the first period, $x \in \{R_1 \text{ (rebuild)}, NR_1 \text{ (not rebuild)}\}$. Owner i chooses x to maximise:

$$V_{1ix} = \pi_{1x} + \beta V_{2|x} + \varepsilon_{1ix},$$

where

$$\pi_{1x} = \begin{cases} (r_1 - c)k_1, & x = R_1 \\ 0, & x = NR_1 \end{cases},$$

V_{1ix} represents the value of choosing x in the first period, β is a discount factor ($0 < \beta < 1$), and ε_{1ix} is an idiosyncratic preference shock for owner i. A value for ε_{1iR_1} could be interpreted, for example, as a randomly drawn quality of constructors, brokers, and renters. On the other hand, a value for ε_{1iNR_1} could be interpreted, for example, as a randomly drawn extent of a tangled title or potential future use of the property by owners or their children. We ignore any other preferences that are common for owner i regardless of their choices, such as utility from holding a piece of land. This is because their expected values are the same across all choices and consequently they cancel out when calculating the probabilities at a later point. We assume that ε_{1ix}, $x \in \{R_1, NR_1\}$, are independently and identically distributed according to the Gumbel distribution. $V_{2|x}$ represents the expected maximum of the value from rebuilding and not rebuilding in the second period conditional on state x, which are given by

$$\begin{aligned} V_{2|R_1} &= \mathrm{E}\left[\max\left\{(r_2 - c)k_2 + \varepsilon_{2iR_2},\ r_2 k_1 + \varepsilon_{2iNR_2}\right\}\right] \\ &= \ln\left(\exp\left((r_2 - c)k_2\right) + \exp\left(r_2 k_1\right)\right) + \gamma, \end{aligned}$$

$$V_{2|NR_1} = \mathrm{E}\left[\max\left\{(r_2 - c)k_2 + \varepsilon_{2iR_2},\ \varepsilon_{2iNR_2}\right\}\right] = \ln\left(\exp\left((r_2 - c)k_2\right) + \exp(0)\right) + \gamma,$$

[7] Building on the dynamic discrete choice model for bus engine replacement by Rust (1987), we apply this framework to the optimal teardowns and redevelopment.

[8] This implicitly assumes that the owners' lifespans continue until the end of period 2.

respectively, where ε_{2iR_2} and ε_{2iNR_2} are idiosyncratic preferences that are independently and identically distributed according to the Gumbel distribution and γ is the Euler's constant.[9] For notational convenience, let

$$A \equiv \exp\left((r_2 - c)k_2\right) + 1, \qquad B \equiv \exp\left((r_2 - c)k_2\right) + \exp\left(r_2 k_1\right),$$

so that $V_{2|NR_1} = \ln A + \gamma$ and $V_{2|R_1} = \ln B + \gamma$. Accordingly, the probability of today's housing dereliction follows a logit model,

$$\Pr(NR_1) = \frac{\exp\left(\bar{V}_{1NR_1}\right)}{\exp\left(\bar{V}_{1R_1}\right) + \exp\left(\bar{V}_{1NR_1}\right)} = \frac{A^\beta}{\exp\left\{(r_1 - c)k_1\right\} B^\beta + A^\beta}, \tag{1}$$

where $\bar{V}$ represents the deterministic component of V and the common term $\exp(\beta\gamma)$ cancels out from the numerator and the denominator. Equivalently, in log-odds form,

$$\ln \frac{\Pr(R_1)}{\Pr(NR_1)} = (r_1 - c)k_1 + \beta \ln \frac{B}{A}. \tag{2}$$

2.2 Future Upzoning

In this subsection, we consider upzoning in the second period — a type of urban regeneration — as a first comparative-statics exercise. In this case, housing owners will be able to rebuild higher buildings due to a regulatory relaxation in the second period. We call the area with this expectation "future upzoning area", in comparison to the "control area" defined as the area with no such expectations. To see the effect of this expectation on today's dereliction probability, we differentiate $\Pr(NR_1)$ with respect to k_2 and obtain $\partial \Pr(NR_1)/\partial k_2 > 0$ if and only if $r_2 > c$.[10]

Intuitively, when rebuilding in the second period is at least marginally profitable, owners in the future upzoning area — expecting an increase in k_2 — are more likely to leave their houses derelict in the first period, because a structure built today would be replaced again once the larger building size becomes available. This means that the expectation of future gains increases today's dereliction probability.

Figure 1 shows a numerical illustration of this exercise, where we set $r_1 = 1$, $k_1 = 1$, $c = 1$, and $\beta = 0.9$. The figure plots $\Pr(NR_1)$ as a function of r_2 for three values of k_2 ($k_2 = k_1, 3k_1, 5k_1$). When $r_2 > c$, the probability of today's dereliction becomes greater as k_2 increases — consistent with the first differentiation result above. Moreover, for sufficiently large k_2 relative to k_1, the dereliction probability is increasing in r_2, meaning that higher expected future profit induces more dereliction today.[11] By contrast, when $k_2 = k_1$, the probability is decreasing in r_2 throughout, since rebuilding in

[9] For detailed derivation, see McFadden (1978).

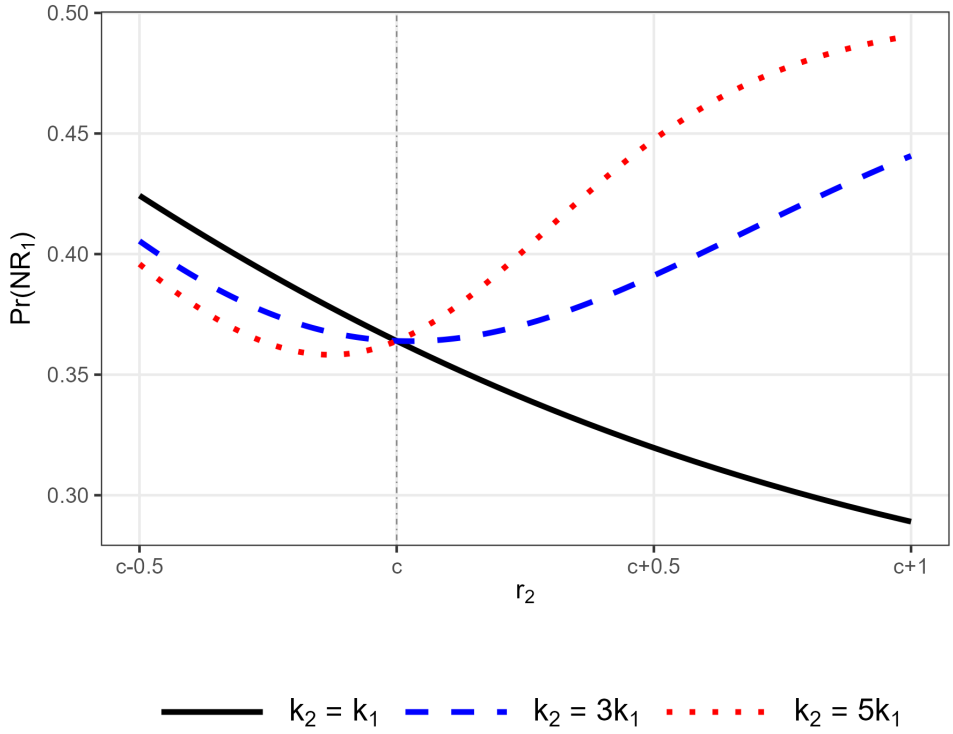
[10] Let $u \equiv \exp\left((r_2 - c)k_2\right)$ and $w \equiv \exp(r_2 k_1)$, and note that $\partial A/\partial k_2 = \partial B/\partial k_2 = u(r_2 - c)$. Differentiating (2) with respect to k_2 gives

$$\frac{\partial}{\partial k_2} \ln \frac{\Pr(R_1)}{\Pr(NR_1)} = \beta\, u\, (r_2 - c) \left(\frac{1}{B} - \frac{1}{A}\right) = -\beta\, (r_2 - c)\, \frac{u\,(w-1)}{AB}.$$

Since $w > 1$, this expression is negative if and only if $r_2 > c$. As $\Pr(NR_1)$ is strictly decreasing in the log odds, it follows that $\partial \Pr(NR_1)/\partial k_2 > 0$ if and only if $r_2 > c$.

[11] Formally, $\partial \Pr(NR_1)/\partial r_2 > 0$ if and only if $k_2 \frac{u}{u+1} \frac{w-1}{w} > k_1$, where u and w are defined as in the preceding footnote.

Figure 1: Future Upzoning and Today's Dereliction Probability



The figure shows a numerical illustration of today's dereliction probability with three scenarios of future upzoning, where we set $r_1 = 1$, $k_1 = 1$, $c = 1$, and $\beta = 0.9$. For $r_2 > c$, the probability of today's dereliction becomes greater as k_2 increases.

the second period yields $(r_2 - c)k_1 < r_2 k_1$ and is therefore strictly dominated by keeping the existing structure. These patterns illustrate how the interaction between future rent levels and permissible building height shapes the rebuilding decision.[12]

2.3 Future Eminent Domain

In this subsection, we consider eminent domain in the second period — another type of urban regeneration — as a second comparative-statics exercise. In this case, housing owners have no choice in the second period, and they are forced to move and get compensation regardless of their first-period choices. We call the area with this expectation "future eminent domain area". Assume that the compensation is assessed at the time the project is announced and is therefore independent of the owner's first-period choice; we denote it by $r_2 k_1$. The dereliction probability is then rewritten as

$$\Pr(NR_1) = \frac{\exp\{\beta r_2 k_1\}}{\exp\{(r_1 - c)k_1 + \beta r_2 k_1\} + \exp\{\beta r_2 k_1\}} = \frac{1}{\exp\{(r_1 - c)k_1\} + 1}. \tag{3}$$

Notably, the dereliction probability in the future eminent domain area depends on neither r_2, k_2, nor

Since both fractions are less than one, $k_2 > k_1$ is necessary but not sufficient.

[12] Figure 1 also shows that, for small values of r_2, all three curves slope downward, reflecting that a decrease in the second-period unit rent discourages owners from rebuilding in either period. For $k_2 = 5k_1$ the curve turns upward already slightly below $r_2 = c$, consistent with the condition in the preceding footnote.

β and is always higher than that of the future upzoning area and control area.[13] Intuitively, the reason is the non-dynamic nature of the future eminent domain area: owners there receive compensation in the second period regardless of their first-period choice. Owners in the future upzoning and control areas, by contrast, lose the option of collecting second-period rent from the structure built today if their first-period choice is "not rebuilding" — this potentially forgone benefit reduces the relative attractiveness of delay.

2.4 Connections to the Empirical Analysis

The model presented in this section underlies the following empirical analysis. Specifically, we first test the theoretical predictions, using future arterial road development — which comes with eminent domain and upzoning — as an arguably exogenous shock. Since arterial road development involves land acquisition on the line of the future road and regulatory relaxation along the adjacent frontage, we assign the former to the future eminent domain area and the latter to the future upzoning area. Subsequently, we exploit this aspect of dereliction occurrence for our instrumental variable estimations and provide evidence of the disamenity effects of derelict houses.

3 Study Area

3.1 Toshima City, Tokyo

The study area for the empirical analysis is Toshima City, one of the 23 Special Wards of Tokyo, Japan. The Special Wards are a form of city-level municipality that were created when the original Tokyo City was abolished in 1943, and they now form the core urban area of contemporary Tokyo. Figure 2 shows the location of Toshima. Among the Special Wards, Toshima is one of the eight central cities of Tokyo, which also include Chiyoda, Minato, Shibuya, Chuo, Shinjuku, Taito, and Bunkyo surrounding the Imperial Palace amid Tokyo's CBD. The centre of Toshima is called Ikebukuro,[14] which is one of Tokyo's three major subcentres — officially designated urban hubs that serve as thriving commercial and transportation centres — along with Shinjuku and Shibuya. The population of Toshima was 295,509 as of March 2026 and its area is 13.01km^2, which is comparable in both population and size to a London Borough.

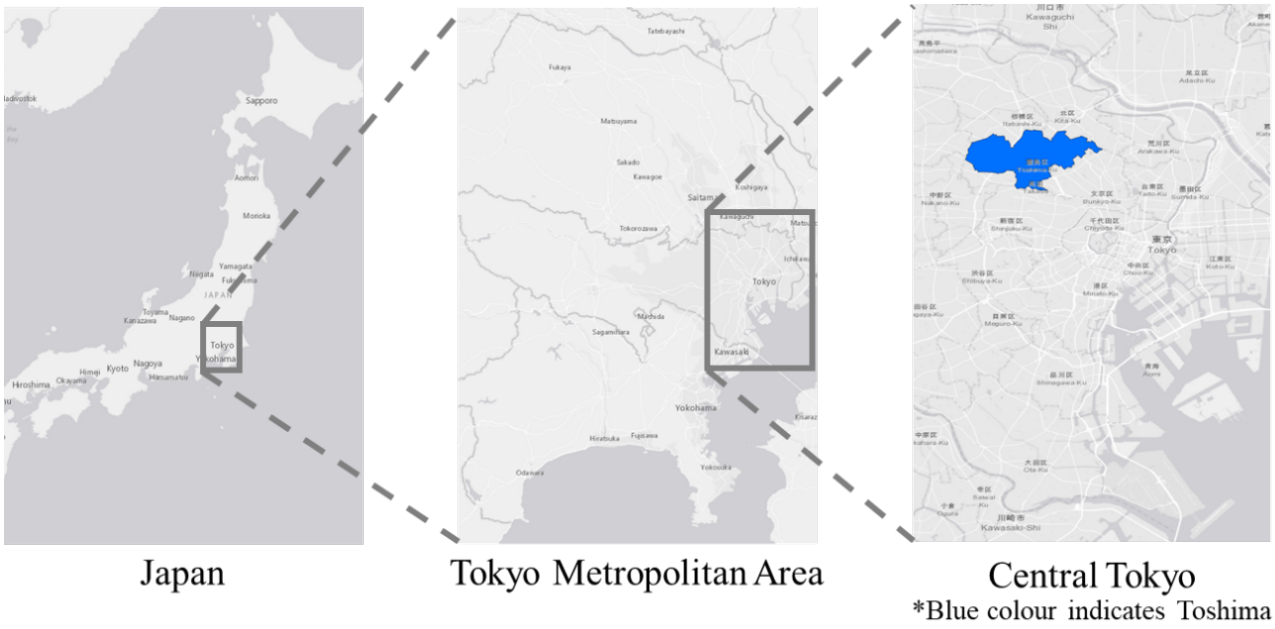
Toshima has ideal conditions for our analysis. First, Toshima exemplifies a prosperous area[15]

[13] In both areas the probability takes the form $\Pr(NR_1) = (1+\Omega)^{-1}$, where Ω denotes the odds of rebuilding. From (3), $\Omega^E = \exp\{(r_1 - c)k_1\}$, whereas from (1), $\Omega^U = \exp\{(r_1 - c)k_1\}(B/A)^\beta$. Since $B > A$ and $\beta > 0$, we have $(B/A)^\beta > 1$ and hence $\Omega^U > \Omega^E$, so that $\Pr(NR_1)$ is strictly higher in the future eminent domain area.

[14] Ikebukuro Station is one of the busiest stations in Tokyo and serves eight railway lines, with approximately 2.3 million daily passengers in 2023.

[15] Despite its prosperity, Toshima had a 4.2% vacancy rate for houses neither for sale nor rent, exceeding the 3.5% average across the 23 Special Wards, according to the Housing and Land Survey in 2018 by the Statistics Bureau of Japan. For reference, Tokyo Prefecture as a whole experienced an annual price increase of 2.9% from 2013 to 2023 and had a 4.2% vacancy rate for houses neither for sale nor rent in 2018. For Osaka City, the core of the second largest metropolitan area of Japan, these figures were 3.0% and 7.6%, respectively. For Nagoya City, the core of the third largest metropolitan area of Japan, these figures were 1.3% and 5.8%, respectively.

Figure 2: Location of the Study Area, Toshima



The study area, Toshima City, is one of the 23 Special Wards (city-level municipalities) of Tokyo. Blue colour indicates Toshima. Among the Special Wards, Toshima is one of the eight central cities of Tokyo. The centre of Toshima is Ikebukuro and is one of the three major subcentres of Tokyo. These mean that Toshima is an example of a thriving area of central Tokyo. The population of Toshima was 295,509 as of 2026, and its area is 13.01km^2, which is comparable in size to a London Borough.

close to Tokyo's central business district. From 2013 to 2023, Toshima's (23 Special Wards') average residential land price increased by 3.97% (3.36%) annually, whereas its housing stock only increased by 1.26% (1.27%) annually, showing the traits of superstar cities.[16] The combination of strong development pressure, the accompanying interest from developers and construction companies, and local governments' financial leeway enables Toshima — in comparison to other ordinary cities in suburban areas — to actively pursue numerous long-run regeneration projects. This offers an adequate superstar-city setting for examining the paradoxical housing dereliction, where a within-city variation in prosperity-driven future expectations plays an important role.

Second, the Toshima city government conducted its own field surveys on neglected empty houses in its jurisdiction, which provide us with parcel-level empty house data. The surveys by Toshima revealed that at least 80% of empty houses in the data were derelict,[17] enabling us to analyse derelict house locations and disamenities.

Third, future arterial road development — as a case of urban regeneration — in Toshima serves as a proxy for sources of today's housing dereliction. Two key factors contribute to this: (1) on the line of the future road, an eminent domain compensation will be provided to housing owners[18] and (2) in

[16]These figures align with Gyourko et al. (2013)'s definition of "superstar cities" (i.e., the sum of the price and housing stock growth rates is above their MSA sample median and the ratio of price growth rate-to-housing unit growth rate is in the top decile), although we only have access to relatively shorter-run trend data.

[17]This is consistent with case in entire Japan. According to the national survey by the Ministry of Land, Infrastructure, Transport and Tourism (MLIT), nearly 90% of empty homes in Japan were constructed before 1980, and those homes are often derelict or not safe to live in due to violations of current building standards.

[18]Explanation of compensation items by Bureau of Urban Development, Tokyo Metropolitan Government is available

areas adjacent to the future road (i.e., within 30m from both edges of the road), land use regulations will be relaxed to encourage dense commercial and residential development after the road construction is completed. Under the Japanese Real Estate Brokerage Act, property brokers must disclose these future plans to homebuyers for case (1), and typically do so in practice for case (2). Consequently, property owners in these areas should be aware of the future road development, expecting eminent domain compensation or regulatory relaxation.

Arterial road plans for the 23 Special Wards, of which those covering Toshima were formulated between 1946 and 1966, were adopted by the Tokyo Metropolitan Government after World War II. The completion rate of the arterial road plans has grown incrementally, with approximately 65% of the plans completed by 2016. On the other hand, the remaining plans have either only received approval but have not yet been built or have not even received approval. This somewhat slow growth trend is partly attributable to the fact that these road plans were designed with a long-run urban construction perspective and were not meant to be completed quickly by construction. Recently, due to the age of the remaining projects, the Tokyo Metropolitan Government has announced a list of remaining projects with a possibility of revision or abolishment, yet no projects in Toshima are on the list.[19] Therefore, the remaining plans in Toshima are expected to be implemented in the future with certainty, shaping current property owners' expectations for the future roads.

3.2 Japanese Urban Cities in General

Japan is experiencing a significant increase in empty homes, primarily driven by demographic factors. This typically occurs through inheritance processes and tangled ownership titles, reflecting Japan's position as one of the world's most rapidly ageing societies (Abe, Fujimaki, Konishi, Usami and Osawa, 2023; Yu and Lee, 2023; Kobayashi, 2016). Many owners who inherited properties from their parents retain empty homes due to various reasons such as the possibility of future use, sentimental value, and preferential property tax treatment.[20] In prosperous urban areas (e.g., Tokyo, Osaka, Nagoya), speculative motives, such as waiting for future property price increases, could provide an additional incentive to retain properties. As a result, owners tend to hold onto these properties without investing in costly maintenance, leading to a widespread problem of neglected and deteriorating houses.

This distinctive context of Japanese housing dereliction provides an ideal research environment for precisely quantifying the disamenity effects of derelict houses on nearby rental markets. Unlike housing abandonment in the United States, which is primarily driven by economic factors such as foreclosures, tax delinquencies, and market value declines (Bender, 1979; White, 1986; Scafidi et al., 1998; Morckel, 2014), Japanese housing dereliction arises from owners' preferences that are largely unrelated to economic distress (Suzuki et al., 2022). This allows us to isolate the pure neighbourhood

at https://www.toshiseibi.metro.tokyo.lg.jp/dainiseibi/jigyou/hosyo_s_02 (retrieved on 7th April 2026).

[19]The list is available at https://www.toshiseibi.metro.tokyo.lg.jp/kotsu_butsuryu/doromou/keikaku_doro/minaoshi (retrieved on 7th April 2026).

[20]Preferential property tax treatment for land plots with dwellings is a well-known factor that disincentivises owners to demolish empty homes despite having little intention to use them. Kobayashi (2016) states that "tax rate on the land is reduced to one sixth of the appraised value if there remains a residential structure on it" and "even after the property is not occupied and had better be demolished, the owner of the property had an incentive to leave the property as it is to enjoy a lower property tax rate".

effects of property dereliction from unobserved neighbourhood distress and confounding effects of foreclosure-driven increases in local housing supply (Anenberg and Kung, 2014). Furthermore, the use of rental data — typical in the Japanese urban economics literature — rather than sales prices enables more accurate estimation of the contemporaneous disamenity effects by avoiding the influence of capitalised expectations about future demolition of derelict properties.

4 Data

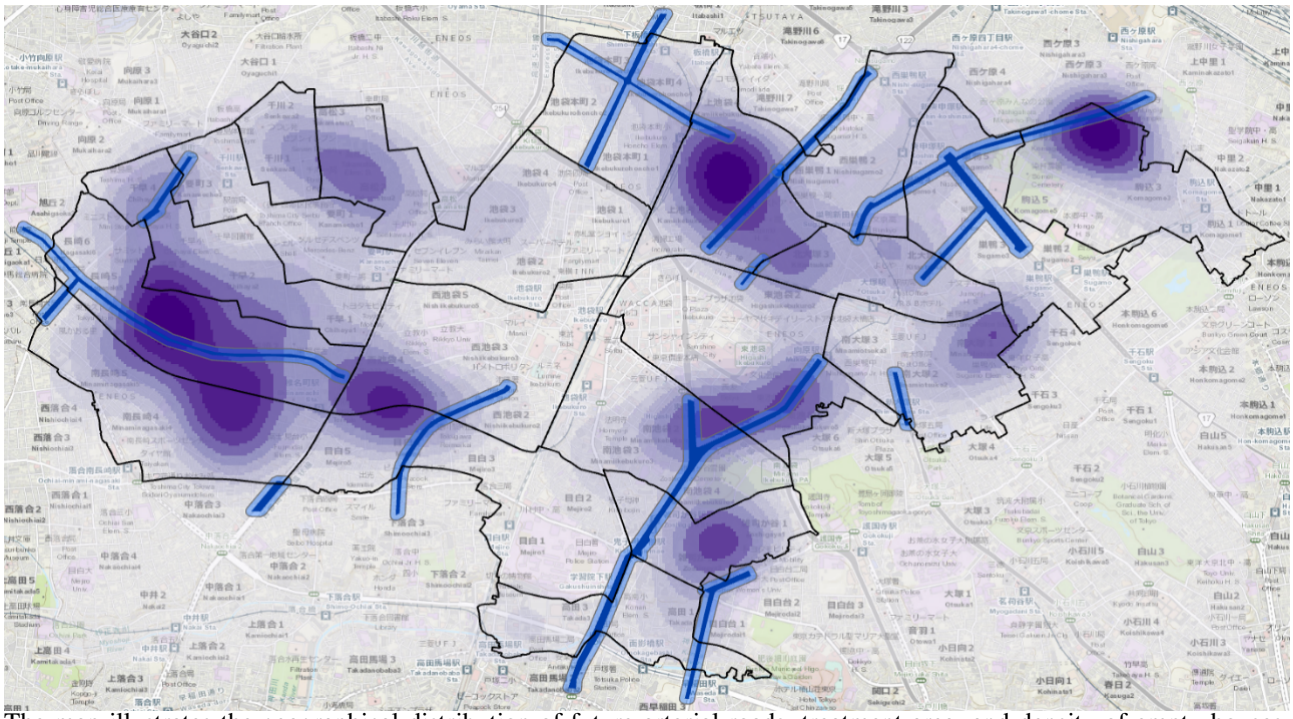
Despite the relatively small area of Toshima City, unique data obtained from field surveys by Toshima city government enable us to conduct important, granular within-city analysis. Toshima city government conducted field surveys on neglected empty houses in its jurisdiction from April 2011 to March 2012 (FY 2011) and from April 2016 to March 2017 (FY 2016). In FY 2011, approximately 20% of entire Toshima was targeted, whereas entire Toshima was targeted in FY 2016. As a result, 367 and 594 empty houses were found through respective surveys. It is important to note that approximately 80% of these empty houses were apparently derelict, showing visible issues such as broken windows or damaged walls. It is also possible that the remaining 20% have some form of dereliction that is not immediately visible from the outside. These surveys provide us with parcel-level data on empty houses, enabling us to pinpoint the geographical location and identify the maintenance condition of each empty house.

Data on future arterial roads (i.e., uncompleted projects) serve as a proxy for owners' future expectations to test our theoretical predictions. These uncompleted projects are shown in Figure 3 as the future arterial roads (i.e., future eminent domain area), and they constitute the treatment area with their 30m buffers (i.e., future upzoning area). The figure also depicts the density of empty houses in Toshima in FY 2016. It appears that empty houses are concentrated in the treatment area (i.e., future eminent domain area + future upzoning area), showing descriptive patterns that align with our theory.

Descriptive statistics provide another piece of evidence. The ratio of derelict houses over sold houses are highest in the future eminent domain area (31.1%), followed by the future upzoning area (30.9%), and lowest in the control area (21.8%), aligning with our theory. The definitions and descriptive statistics of all variables are shown in Appendix A.

For the estimation of the causes of housing dereliction (dereliction analysis), we use the empty house data from FY 2016 provided by Toshima city government, parcel-level property transaction data from 2010 to 2018 provided by MLIT, and the future arterial road data. The unit of observation is the individual house. Because the transaction data only go back to 2010, we forgo the use of empty house data in FY2011 and focus on comparing empty houses in FY 2016 with transacted houses from 2010 to 2018. This allows us to construct a sample of empty (derelict) and transacted houses that are comparable in terms of the timing of when their owners vacated them. Despite the lack of vacancy duration information in our study area, we believe that this comparison is appropriate, as we are aware that a median of vacancy duration is around 5 years in another city in the Tokyo metropolitan area (Suzuki and Hino, 2018). Additionally, we use GIS data from the MLIT National Land Numerical

Figure 3: Future Arterial Roads, Treatment Area, and Density of Empty Houses



The map illustrates the geographical distribution of future arterial roads, treatment area, and density of empty houses, which are often derelict. Dark navy features represent the future roads. These features (future eminent domain area) and their 30m buffers (future upzoning area) constitute the treatment area in the empirical analysis. Kernel density of empty houses in FY 2016 is shown by purple shades. Darker shades of purple indicate a higher density of empty houses. It appears that empty houses are concentrated in the treatment area. Black lines delineate the borders of neighbourhoods in Toshima. Each neighbourhood has a population of approximately 12,800 and an area of approximately 0.65 km^2 (i.e., 800m×800m) on average.

Information[21] to control for other factors that may correlate with housing dereliction and future road locations.

For the estimation of the disamenity effects (rent analysis), we use the empty house data in FY 2011 and FY 2016 and rental apartment unit listings from 2011 to 2012 and from 2016 to 2017. Additionally, the future arterial road data are used to predict empty house locations. The unit of observation is the individual rental apartment unit, and we estimate the effect of proximity to empty (derelict) houses on rents. The rental data are provided by a Japanese company, Mansion Research Ltd.,[22] whose business activities include consulting services for property brokers and the provision of real estate information to consumers. Parcel-level locations, housing attributes, advertised rents, and listing dates of each rental unit are included in the data. We then count the number of nearby empty houses and calculate the coverage of the treatment area around each rental unit. We also use comprehensive building data from Zenrin Zmap-TOWNII[23] to control for the number of nearby houses. Moreover, GIS data from the MLIT National Land Numerical Information and open data provided by Toshima city government

[21] Publicly available GIS data provided by MLIT are available at https://nlftp.mlit.go.jp/ksj/ (retrieved on 7th April 2026).

[22] In Japanese, the term "mansion" (*manshon*) refers to an apartment building, rather than a large house as used in English.

[23] Zenrin Zmap-TOWNII is residential map GIS data released by Zenrin, a Japanese map publishing company. The data were provided to the authors by the Center for Spatial Information Science (CSIS) at the University of Tokyo.

are used to control for other factors that may correlate with rents, housing dereliction, and future road locations. The definitions of variables and descriptive statistics are shown in Appendix A.

5 Empirical Model

5.1 Dereliction Analysis: Effects of Future Expectations on Housing Dereliction

To validate the descriptive evidence presented in the previous section, we apply logit models to parcel-level data on houses whose owners have recently vacated them (i.e., empty or derelict houses and sold houses). The models evaluate the effects of expectations regarding future eminent domain and upzoning on today's owners' decisions to neglect their houses after controlling for other factors. One may be concerned that policymakers tend to plan the arterial roads in areas with high housing dereliction rates, which potentially raises an endogeneity concern. However, since the road plan decisions were completed by 1966, it is highly unlikely that policymakers of that era could have anticipated today's dereliction patterns. Thus, we can rule out the potential reverse causality. Moreover, we control for neighbourhood-level fixed effects[24] and housing and neighbourhood attributes that may simultaneously correlate with housing dereliction and road plans, thereby addressing potential omitted variable bias. Furthermore, we have no evidence of partial correlations between future road locations and block-level attributes within each neighbourhood, which further reduces concerns about omitted variable bias.

We compare empty houses — which are often derelict — with sold houses and examine whether the future eminent domain and upzoning affect the probability of houses to become empty or derelict after owners' exit. An advantage of this comparison is the ability to control for detailed housing attributes, which are not available for occupied houses. Our baseline model is then defined as follows:

$$Emp_i^* = \alpha_1 + \beta_1 Treat_i + X_i\gamma_1 + \delta_n + \varepsilon_i,$$

$$Emp_i = \begin{cases} 1 & \text{if } Emp_i^* > 0 \\ 0 & \text{if } Emp_i^* \leq 0 \end{cases},$$

$$\Pr\left(Emp_i = 1 \mid Treat_i, X_i, \delta_n\right) = \Lambda\left\{\alpha_1 + \beta_1 Treat_i + X_i\gamma_1 + \delta_n\right\},$$

where Emp_i^* is a latent variable that represents the difference between a housing owner's utility from keeping the house empty and that from selling the house. The owner keeps it empty ($Emp_i = 1$) if the latent variable takes a positive value and sells it ($Emp_i = 0$) if the latent variable takes a negative value. $Treat_i$ is a dummy variable that takes the value of one if house i's footprint is in the treatment area and takes the value of zero if house i is located elsewhere in Toshima. We also estimate the model where $Treat_i$ is decomposed into $EminentDomain_i$, a dummy variable that takes

[24] There are 20 neighbourhoods in Toshima. Each neighbourhood has a population of approximately 12,800 and an area of approximately 0.65 km^2, i.e., 800m×800m, on average.

the value of one if house i's footprint is in the future eminent domain area, and $Upzoning_i$, a dummy variable that takes the value of one if house i's footprint is in the future upzoning area, to test our two theoretical predictions separately. X_i is a row vector of control variables representing house i's attributes and neighbourhood characteristics. α_1 and β_1 are parameters, γ_1 is a column vector of parameters, δ_n represents neighbourhood-level fixed effects, and ε_i is an error term that follows the logistic distribution. Λ represents the logistic function.

Additionally, we compare derelict houses and well-maintained vacant houses, respectively, against sold houses. For this purpose, Emp_i^* is replaced with Der_i^* and $NonDer_i^*$, respectively. The variable Der_i^* is a latent variable that represents the difference between a housing owner's utility from keeping the house derelict and that from selling the house. The owner keeps it derelict ($Der_i = 1$) if the latent variable takes a positive value and sells it ($Der_i = 0$) if the latent variable takes a negative value. Similarly, the variable $NonDer_i^*$ is a latent variable that represents the difference between a housing owner's utility from keeping the house vacant in good condition and that from selling the house. The owner keeps it vacant in good condition ($NonDer_i = 1$) if the latent variable takes a positive value and sells it ($NonDer_i = 0$) if the latent variable takes a negative value.

5.2 Rent Analysis: Effects of Derelict Houses on Nearby Rents

Regarding the disamenity effects of derelict houses (rent analysis), we estimate the impact of empty houses — which are often derelict — on nearby rents using a hedonic model. The unit of observation here is the individual rental apartment unit. Let $Rent_{ft}$ be the monthly rent of rental apartment unit f in year $t \in \{2011, 2016\}$. $Empty_{rft}$ is the number of empty (derelict) houses within rm from a rental apartment unit f in year t. X_{ft} is a row vector of control variables representing unit f's attributes and its neighbourhood characteristics. In addition, we control for unobservable location attractiveness using neighbourhood-year fixed effects and an indicator for nearby apartment buildings with unusually high vacancy rates.[25]

Even after controlling for the factors that may affect rents and nearby empty house occurrences, endogeneity concerns still remain. For example, houses in low-priced areas may be more likely to become derelict due to unobservable unattractiveness, which standard control variables may not fully capture. Conversely, houses in high-priced areas may also be more likely to remain unattended or derelict, as owners may have high reservation prices (Kanayama and Sadayuki, 2021) and anticipate further profit increases due to inelastic supply. Moreover, a potential reverse causality is also a concern. For instance, surrounding property rents may act as market activity signals for owners of empty houses and potentially influence their reutilisation decisions. Specifically, strong price appreciation could discourage owners from selling (Anas and Eum, 1984), potentially contributing to current vacancy or

[25]Neighbourhood-year fixed effects are necessary but may not fully address the endogeneity concern. Therefore, we replicate a variable regarding nearby apartment buildings with unusually high vacancy rates (i.e., over 30%), which were used in Sadayuki et al. (2020), to control for further confounding factors arising from the popularity of each location. An apartment building with a high vacancy rate may involve a slight disamenity effect similar to a derelict house, however, an apartment building is more likely to be well-maintained due to the presence of a certain number of occupied units, according to the field survey report by the Toshima city government. Thus, we assume that the proximity to an apartment building with a high vacancy rate mainly represents the attractiveness of respective areas at a granular level.

dereliction.

To address the endogeneity concerns, we employ an instrumental variable strategy, based on our theoretical prediction. Our theory predicts that owners of derelict houses in the treatment area are more likely to leave their properties derelict. Thus, it is natural to employ the ratio of the treatment area within a circle of radius r around unit f, $TreatArea_{rf}$, as the instrumental variable.[26] Then, our empirical model to estimate the disamenity effect can be described as follows:

$$\ln Rent_{ft} = \alpha_2 + \beta_2 \widehat{\boldsymbol{Empty}}_{rft} + X_{ft}\gamma_2 + \delta_{nt} + \varepsilon_{ft},$$

$$\boldsymbol{Empty}_{rft} = \tilde{\alpha}_2 + \tilde{\beta}_2 TreatArea_{rf} + X_{ft}\tilde{\gamma}_2 + \tilde{\delta}_{nt} + \tilde{\varepsilon}_{ft},$$

where α_2 and β_2 are parameters, and γ_2 is a column vector of parameters associated with a row vector of control variables X_{ft}. Neighbourhood-year fixed effects are represented by δ_{nt}, and ε_{ft} is an error term. Bold indicates that the number of empty (derelict) houses $Empty_{rft}$ is instrumented by the treatment area coverage. Note that the $\sim$ refers to first-stage parameters. We estimate this model by applying 2SLS. Following Hartley (2014), β_2 in our model should represent the disamenity effect of empty (derelict) houses without confounding supply effects.[27]

The instrumental variable needs to strongly correlate with the endogenous variable. Our theory suggests that owners are more likely to neglect their properties in the treatment area, supporting the relevance of the instrumental variable. As shown in Figure 4, we expect that greater treatment area coverage within a given radius r increases the likelihood of derelict house occurrence within that radius. If this holds empirically, $TreatArea_{rf}$ is an appropriate variable to predict $Empty_{rft}$ for every radius r.

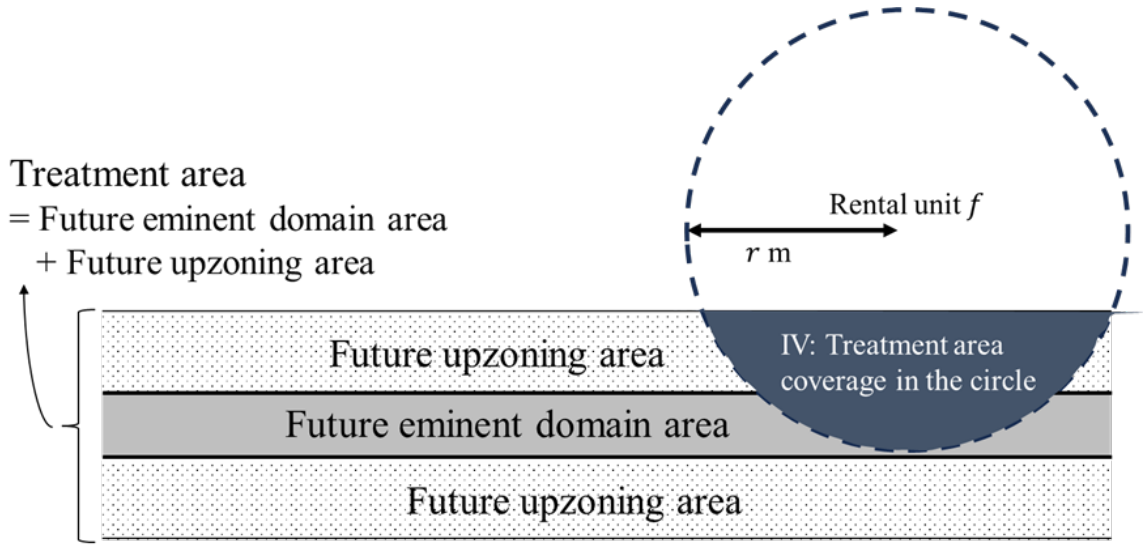
On the other hand, the instrumental variable must not correlate with the error term ε_{ft}. We believe that the exclusion restriction is satisfied in our estimation for the following reasons. First, future arterial roads provide no amenities or disamenities before construction, which is not expected to take place in the near future. Thus, the proximity to future roads should not be capitalised into today's rents, which are determined by two-year contracts and reflect the value of housing services in specific periods before the road construction. Indeed, by exploiting a case of transport improvement in Tokyo, we find an increase in rents only after the new transport was put into service (see Appendix B). Additionally, the absence of rent control — which would otherwise incentivise landlords to gradually raise rents in anticipation of future rent increase limits — in the Japanese housing market ensures that the future expectations are not capitalised into current rents.

Second, omitted variables that may simultaneously correlate with rents and future road locations are adequately addressed in our estimation. The potential omitted variables are not of concern to some extent, since the road plan decisions were completed by 1966, and thus it is highly unlikely

[26] One may argue that using two instruments, which are the ratio of the future eminent domain area and that of the future upzoning area, would be more appropriate. However, they are highly correlated ($\rho \approx 0.9$). Thus, to avoid multicollinearity, we use $TreatArea_{rf}$ as the single instrument in the main specification of the disamenity effect analysis. In the robustness check, we also use the two instruments (see Appendix C).

[27] Using the Japanese housing market data has other advantages. First, the absence of the contaminating effects of foreclosure (Suzuki et al., 2022) allows us to purely estimate the disamenity effects. Second, as our data only contain empty houses neither for sale nor rent, the effect of derelict houses is not confounded by the effect of excess housing supply.

Figure 4: Illustration of the Instrumental Variable



The figure illustrates the idea of our instrumental variable. Treatment area consists of future eminent domain area and future upzoning area. Future eminent domain area is empirically defined as areas on the line of future roads. Future upzoning area is empirically defined as areas adjacent to (i.e., within 30m from both edges of) future roads. The larger the coverage of the treatment area in the circle of radius rm, the higher the probability of derelict houses being in the circle, thus, treatment area coverage in the circle predicts the number of derelict houses in the circle.

that policymakers of that era could have anticipated current housing dereliction and other micro-level demographic patterns. To further ensure the exogeneity of the instrument, we control for factors that may simultaneously affect rents and future roads, such as (1) neighbourhood-year fixed effects and (2) proximity to existing arterial roads. These control variables respectively address potential concerns that (1) roads may have been more likely to be planned in originally low-demand neighbourhoods with lower rents or implementation of roads may be more likely to be prioritised in economically growing neighbourhoods with higher rents and (2) areas with future roads may have low demand and lower rents due to poor accessibility to existing arterial roads. Regarding the former, one may still be concerned that, within each neighbourhood, future roads may have been planned or more likely to be postponed in areas that are poor, less demanded, commercially insignificant, or have strong community ties. Reassuringly, our placebo tests show no partial correlation between block-level attributes that represent poverty, demand, commercial significance, and community ties and future road locations, conditional on neighbourhood-level fixed effects (see Section 6.1).

Third, we have no evidence showing that future roads affect rents through other factors rather than nearby housing dereliction. Potentially, the future expectations may induce (1) maintenance neglect of rental apartment units, (2) nearby redevelopment, and (3) changes in time-on-market, all of which could be exclusion channels. To show the first concern does not affect our results, we re-estimate the empirical model after excluding the rental apartment units located in the treatment area — which are potentially neglected and thus valued less — and confirm that the results remain largely unchanged (see Appendix C).[28] Regarding the second and the third concerns, we run placebo first-stage specifications.

[28]We are also aware that the rental units used in our estimation are likely to be well-maintained even in areas on the line of or near future roads, as they are advertised on the market and owners would maintain the property for rental use and to uphold their reputation with other tenants in the same building. In fact, according to the empty houses field survey report

If the future expectations affect rents through stimulating nearby redevelopment or changing time-on-market, we would obtain statistically significant results in the placebo first-stage estimations, where we replace the number of empty houses with the number of nearby new buildings and time-on-market of each rental apartment unit, respectively. We then find no significant relationships, showing that the potential exclusion via nearby redevelopment and time-on-market are not necessarily the case (see Appendix F).

Controlling for all conceivable factors and employing an instrumental variable strategy help address potential endogeneity concerns that the previous literature has not fully resolved. Using this novel empirical strategy, we quantify the disamenity effect in the following section. We also examine its heterogeneity depending on accessibility to public safety services, which may alleviate the negative impacts of derelict houses.

6 Empirical Results

6.1 Future Road Locations

Before going into our main estimation results, we show that future road locations are as good as random conditional on neighbourhood fixed effects. One may be concerned that, within each neighbourhood, future roads may have been planned or more likely to be postponed in blocks that are poor, commercially insignificant, or have strong community ties. For instance, policymakers may have chosen historically poor blocks as the future road sites, and such poverty, including low rents, may persist until today (Heblich, Trew and Zylberberg, 2021). Conversely, road plans in high-demand areas such as city centre or commercial areas may have been prioritised and thus less likely to remain uncompleted in such high-priced blocks today. Furthermore, stronger community ties may contribute to influentially oppose future road constructions and result in higher probability of leaving road plans uncompleted.[29] At the same time, community activities may also contribute to enhance safe living environment and raise rents through activities such as crime watch, threatening the exogeneity of future road locations.

There are 83 blocks in 20 neighbourhoods in our study area, Toshima. We use block-level data in 2016 to show that the abovementioned concerns are not the case. Regarding poverty, as income level or share of low-skilled workers are not available, we use average housing size as a proxy for wealth of each block. Commercial significance of each block is proxied by the distance to the city centre and nearest commercial area. As a proxy for the strength of community ties, we use participation

in 2016 by the Toshima City Government, we find that 184 out of 229 apartment buildings were not derelict, even among apartment buildings with unusually high vacancy rates (i.e., over 30%). This is in contrast with empty houses, where 445 out of 594 houses had some form of dereliction. Given that these facts apply to apartment buildings with unusually high vacancy rates, we can reasonably expect that advertised properties, which may not necessarily have high vacancy rates, are even better maintained. That said, apartment units near future roads may still tend to be lower-priced due to less refurbishment and/or fixed-term contracts, which allow owners to refuse contract renewals unlike ordinary contracts that typically protect tenants. To address these potential concerns, we control for the refurbishment status of each rental unit and the contract type in our estimation.

[29] In the case of the US Interstate Highways constructed in 1950s and 1960s, it is known that poor (blight) areas were targeted for removal, and it is also known that local communities successfully prevented road construction in Baltimore (Mohl, 2004).

Table 1: Block-Level Placebo Tests

	(1) OLS	(2) OLS	(3) OLS	(4) OLS	(5) OLS	(6) OLS
Dependent Variable	HouseSize	DistCom	DistCentre	Community	ShOver20yrs	ShInMigrants
FutureRoad	-6.207	18.46	119.5	-0.0097	0.0249	-0.0118
	(4.410)	(15.70)	(122.6)	(1.300)	(0.0191)	(0.0145)
Neighbourhood FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.083	0.385	0.800	0.737	0.032	0.117
Observations	83	83	83	83	83	83

Notes: ***, **, and * indicate 1 percent, 5 percent, and 10 percent significance levels, respectively. No statistically significant relationships were found (i.e., future roads are not necessarily located in blocks that are poor or commercially insignificant or have strong community ties). Heteroskedasticity-robust standard errors are in parentheses. Full results are available upon request.

ratio of neighbourhood associations (Pekkanen, Tsujinaka and Yamamoto, 2014). Another proxy for community ties is the share of residents who have lived in the same block for more than 20 years and that of in-migrants in the past five years, given that inflow of new residents dilutes social capital in each area (Hilber, 2010). Definitions of variables and descriptive statistics are shown in Appendix A.

We estimate the following model to ensure that there are no statistically significant relationships between the block-level attributes and future road locations:

$$Block_b = \alpha_3 + \beta_3 FutureRoad_b + \delta_n + \varepsilon_b,$$

where $Block_b$ is an attribute of block b in or around 2016 (subject to data availability), and $Block_b \in \{HouseSize_b, DistCentre_b, DistCom_b, Community_b, ShOver20yrs_b, ShInMigrants_b\}$. $FutureRoad_b$ is the dummy variable that takes the value of one if there is a future road in block b in 2016. δ_n is the neighbourhood-level fixed effects, α_3 and β_3 are parameters, ε_b is the error term.

Table 1 presents the estimation results. Reassuringly, no partial correlation between the block-level attributes and future road locations are found conditional on neighbourhood fixed effects. We can interpret the results to mean that future roads are not necessarily located in blocks that are poor or commercially insignificant or have strong community ties.

6.2 Dereliction Analysis: Effects of Future Expectations on Housing Dereliction

Table 2 presents the empirical results of dereliction analysis using logit models, where houses are classified as treated if their footprint is located in either the future eminent domain area or the future upzoning area.[30] Each column reports the average partial effect of the treatment. Column (1) presents

[30] A large-scale property development, which acquires land or existing property, can also be expected to have similar effects on empty housing occurrence as in the future eminent domain area. However, as of 2016, such developments were only planned in limited areas around major stations, where there are very few houses to be vacant. In contrast, road plans exist citywide, providing rich within-city variation. We therefore focus on road developments in this paper.

Table 2: Effects of Future Urban Regeneration on Housing Dereliction

	(1) Logit	(2) Logit	(3) Logit	(4) Logit	(5) Logit	(6) Logit
Dependent Variable	Vacant	Derelict	Vacant, not derelict	Vacant	Derelict	Vacant, not derelict
Treat	0.076**	0.090***	-0.006	–	–	–
	(0.034)	(0.032)	(0.028)	–	–	–
EminentDomain	–	–	–	0.108	0.157**	-0.055
	–	–	–	(0.067)	(0.065)	(0.051)
UpZoning	–	–	–	0.068*	0.072**	0.005
	–	–	–	(0.037)	(0.035)	(0.032)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Neighbourhood FE	Yes	Yes	Yes	Yes	Yes	Yes
Log Likelihood	-961.442	-789.420	-403.723	-961.300	-788.748	-403.196
Observations	1961	1812	1516	1961	1812	1516

Notes: ***, **, and * indicate 1 percent, 5 percent, and 10 percent significance levels, respectively. Average partial effects are reported. Treatment consists of eminent domain and upzoning, which are mutually exclusive. The omitted category is properties outside the treatment area. Bootstrapped standard errors in parentheses (1,000 replications). Columns (1) and (4) use the full sample. Columns (2) and (5) use Derelict and Sold sample. Columns (3) and (6) use Vacant, not derelict, and Sold sample. Full results are available upon request.

the average partial effect of the treatment on today's housing vacancy, where we compare empty houses — including derelict houses — against sold houses. Columns (2) and (3) compare derelict houses (i.e., empty houses in bad condition) and and other empty houses (i.e., empty houses in good condition), respectively, against sold houses. The results indicate that the treatment is associated with a 7.6-percentage-point increase in the probability of vacancy after owners' exit, and this is mostly driven by the effect on the dereliction probability — a 9.0-percentage-point increase. In contrast, the treatment has no statistically significant effect on the occurrence of well-maintained empty houses.

Columns (4) to (6) show the results of the dereliction analysis with separately defined two types of treatment. Each column corresponds to the previous columns, reporting the average partial effect of each type of treatment — future eminent domain and upzoning — on the probability of vacancy, dereliction, and well-maintained vacancy. The results indicate that being located in the future eminent domain area is associated with a 15.7-percentage-point increase in the probability of dereliction after owners' exit, whereas being located in the future upzoning area is associated with a more modest 7.2-percentage-point increase, aligning with the model predictions. It is also worth mentioning that the results consistently show no statistically significant effects of the future eminent domain and upzoning on the probability of well-maintained vacancy.

Overall, the empirical results are consistent with our theory: future expectations incentivise owners of derelict houses to delay rebuilding their properties. This restricts effective housing supply in a high-demand market and can be considered as a side effect of a place-based urban regeneration policy. Moreover, our findings indicate that these expectations predict the location of derelict houses, but not of well-maintained empty houses, which means that the local average treatment effect estimated

Table 3: Effects of Derelict Houses on Nearby Rents

	(1)		(2)		(3)	
	OLS	2SLS	OLS	2SLS	OLS	2SLS
Dependent Variable	ln(Rent)	ln(Rent)	ln(Rent)	ln(Rent)	ln(Rent)	ln(Rent)
Second stage						
Empty_{80}	-0.0032***	**-0.0163****	-0.0039***	**-0.0370*****	-0.0028***	**-0.0671*****
	(0.0006)	**(0.0083)**	(0.0008)	**(0.0125)**	(0.0009)	**(0.0237)**
First stage						
TreatArea_{80}	–	0.1381***	–	0.1499***	–	0.1005***
	–	(0.0201)	–	(0.0274)	–	(0.0242)
Sample	Full Sample		Subsample (Distant from Fire Stations)		Subsample (Distant from Police Stations)	
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Neighbourhood FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.892	–	0.888	–	0.876	–
KP F-statistic	–	47.4	–	29.9	–	17.2
Observations	13936	13936	7136	7136	6739	6739

The unit of observation is the individual rental apartment unit. The dependent variable of the second stage is the logarithmic value of the monthly rent of a rental apartment unit. ***, **, and * indicate 1 percent, 5 percent, and 10 percent significance levels, respectively. Heteroskedasticity-robust standard errors are in parentheses. **Bold** indicates that Empty_{80} is instrumented. Full results are available upon request.

in the following hedonic section represents the impact of derelict houses on their neighbourhoods.

6.3 Rent Analysis: Effects of Derelict Houses on Nearby Rents

Table 3 presents empirical results of rent analysis. We use $r = 80$ for our main specification, as this results in the strongest first-stage estimation. Columns (1), (2), and (3) show the results of the full sample analysis as our baseline and subsample analysis using observations distant from public safety facilities, respectively. The variable of interest is $Empty$ (i.e., the number of nearby empty houses), conditional on the total number of nearby houses.

The first-stage results show that one standard deviation increase in the treatment area coverage in a circle of radius 80m is significantly associated with an approximately 0.10 to 0.15 units increase in the number of empty houses in the circle, ensuring the relevance of our instrumental variable.[31] Looking back to the empirical results of dereliction analysis, we can interpret that these empty houses are likely to be derelict.

While OLS results show relatively small magnitude of the effect of one additional derelict house on nearby rents, 2SLS results indicate much larger impacts. In other words, OLS estimates are biased upward. This can be interpreted that derelict houses in the treatment area are more likely to be located in areas with unobservable factors that contribute to raise rents. Underlying mechanisms would include owners' high reservation prices in areas with expectations of large-scale redevelopment (Kanayama

[31]The 80-metre radius is mostly equivalent to the one (0.05 miles) used in Paredes and Skidmore (2017) and is our preferred radius due to the strongest first stage.

and Sadayuki, 2021), an expectation on a further price appreciation due to inelastic housing supply, and a strong price appreciation stimulating owners not to immediately sell their houses (Anas and Eum, 1984).

Second-stage results from column (1) show that, overall, the effect of derelict houses is statistically and economically significant, implying that one additional derelict house decreases nearby rents by approximately 1.6%. We also re-run this baseline estimation after replacing $Empty$ with $Derelict$ defined in Appendix A to see if the estimated disamenity effect is robust to the definition of the endogenous variable and find that the results remain largely unchanged (for estimation results, see Appendix C). Moreover, for another robustness check, we re-estimate the model with various radii r and confirm that the implications remain unchanged regardless of the value of r. Estimated effects diminish as r increases (see Appendix D), which is plausible as, for example, potential renters would care less about derelict houses 100m away than those 60m away.

Interestingly, columns (2) and (3) show that the estimated disamenity effects are pronounced in areas distant from fire stations or police stations,[32] showing that one additional derelict house decreases nearby rents by approximately 3.6% and 6.5% in respective areas. These findings align with the literature that indicates derelict houses attract fire and crime (Ellen et al., 2013; Chang and Jacobson, 2017), and fear of crime lowers nearby property values (Gibbons, 2004). Given this literature, our results can be interpreted that low accessibility to the public safety services amplifies potential renters' concerns about higher risks of fire and crime attributable to derelict houses.

Given that the accessibility to fire stations or police stations matters, the disamenity effects of derelict houses could be muted by proximity to these facilities. We test this mirror image by using the other subsamples — rental apartment units near fire stations or police stations.[33] The results are in Table 4, showing no significant effects of derelict houses on rents in areas near these public safety services.

One may be concerned that the low accessibility to public safety services may represent other confounding factors, such as peripheral locations within neighbourhoods. To support our findings, we also conduct another subsample analysis to show that low accessibility to placebo public facilities — which are not expected to alleviate the disamenity effects — does not amplify the effects of derelict houses on rents. The results are in Appendix E.

7 Conclusions

This paper sheds light on derelict houses in a superstar city — which are at least partly induced by future expectations — and quantifies the disamenity effect of derelict houses. We first develop a dynamic discrete choice model to understand how future expectations prolong dereliction status, focusing on urban regeneration prospects. The model predicts that the dereliction probability is

[32] When distance from rental apartment unit f to closest facility g is above mean, the rental apartment unit f is categorised into the subsample consisting of observations distant from g. This definition applies throughout the paper.

[33] When distance from rental apartment unit f to closest facility g is less than mean, the rental apartment unit f is categorised into the subsample consisting of observations near g. This definition applies throughout the paper.

Table 4: Effects of Derelict Houses on Nearby Rents Near Public Safety Services

	(1)		(2)	
	OLS	2SLS	OLS	2SLS
Dep.Var.	ln(Rent)	ln(Rent)	ln(Rent)	ln(Rent)
Second stage				
Empty_{80}	-0.0010	**0.0109**	-0.0036***	**0.0034**
	(0.0009)	**(0.0082)**	(0.0008)	**(0.0080)**
First stage				
TreatArea_{80}	–	0.2071***	–	0.1922***
	–	(0.0288)	–	(0.0302)
Sample	Subsample (Near Fire Stations)		Subsample (Near Police Stations)	
Control Variables	Yes	Yes	Yes	Yes
Neighbourhood FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Adjusted R^2	0.898	–	0.908	–
KP F-statistic	–	51.7	–	40.5
Observations	6800	6800	7195	7195

The unit of observation is the individual rental apartment unit. The dependent variable of the second stage is the logarithmic value of the monthly rent of a rental apartment unit. ***, **, and * indicate 1 percent, 5 percent, and 10 percent significance levels, respectively. Heteroskedasticity-robust standard errors are in parentheses. **Bold** indicates that Empty_{80} is instrumented. Full results are available upon request.

higher when future urban regeneration expectations exist. Our theory provides economic rationality at a granular within-city level for owners to strategically wait for the future increased profits despite current high housing demand, filling the gap in more aggregated analysis in the previous literature.

We subsequently provide empirical evidence of this mechanism using unique parcel-level data from central Tokyo. Results show that the expectations of future urban regeneration are associated with an approximately 7- to 16-percentage-point increase in the probability of housing dereliction. Importantly, the empirical results show that those future expectations are not associated with housing vacancy in good condition, motivating the following estimation of the disamenity effects.

Finally, by leveraging these insights as a novel instrument to overcome methodological challenges that the literature has faced, we show economically and statistically significant disamenity effects of derelict houses. Overall, the empirical result shows that one additional derelict house reduces nearby rents by approximately 1.6%, implying a social welfare loss through a disamenity channel. The subsample analysis shows approximately 3.6% and 6.5% rent declines in areas distant from fire stations and police stations, respectively. These results imply that the potential renters' concerns about higher risks of fire and crime attributable to derelict houses may be further amplified due to low accessibility to the public safety services.

Given that national and local governments are responsible for taking countermeasures against derelict houses (Sadayuki et al., 2020; Spader et al., 2016; Paredes and Skidmore, 2017; Segú, 2020; Biswas et al., 2021), understanding the mechanisms of derelict house occurrences and assessing the

negative impacts of derelict houses are essential for evidence-based policymaking. First, understanding why owners leave their properties derelict despite strong housing demand is of paramount importance for facilitating housing supply. Governments should minimise unintended adverse effects of future urban policies on underutilisation of today's real estate by incentivising, for example, temporary (immediate) utilisation of the properties in the future eminent domain (upzoning) area. Second, quantifying the disamenity effects of derelict houses is also crucial for ensuring accountability to taxpayers. This helps governments secure budgets for improving the living environment by reducing derelict houses, which may even lead to an increase in property tax revenue (Sadayuki et al., 2020). Third, this paper reveals previously overlooked benefits of public safety services. Given that some local governments have been discussing consolidation of public safety facilities, our findings should be considered in evaluating such policies.

It is worth noting the limitations of this study. First, while future urban regeneration is one significant factor that explains the occurrence of derelict houses in a superstar city, there are still many factors that could contribute to housing dereliction. Taking these factors into account would lead to a comprehensive formalisation of derelict house occurrences. Second, this comprehensive formalisation may help capture more variation in the occurrence of derelict houses, further improving the disamenity effect estimations. Third, estimating the effects of heterogenous maintenance conditions of derelict houses by extending our methodology would further deepen the discussion.

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Appendix for

"Housing Dereliction in a Superstar City: Speculative Neglect and Its Disamenity Effects"

A Descriptive Statistics

Table A.1: Definitions of Variables and Descriptive Statistics for Block-Level Placebo Test

Variable	Definition	Mean	Std. dev.	Min	Max
Dependent variables					
HouseSize	Average footprint of houses (square metres)	58.7	15.5	24.1	157.0
DistCom	Distance from the centroid of block to nearest commercial area (metres)	81.3	77.0	0	327.0
DistCentre	Distance from the centre of block to the city centre (metres)	1662.0	753.0	75.1	3529.0
Community	Participation rate of neighbourhood associations	50.5	9.75	40.5	68.5
ShOver20yrs	Share of residents who have lived in the same block for more than 20 years	0.244	0.0655	0.0280	0.354
ShInMigrants	Share of in-migrants in the last five years	0.207	0.0476	0.114	0.386
Independent variables					
FutureRoad	1 = future road exists in block; 0 = o.w.	0.614	0.490	0	1
Neighbourhood_n	1 = block located in neighbourhood n (20 types); 0 = o.w.	–	–	0	1

Observations = 83

Table A.2: Definitions of Variables and Descriptive Statistics for Dereliction Analysis

Variable	Definition	Mean	Std. dev.	Min	Max
Future Eminent Domain Area					
Dependent variables					
Emp	1 = empty house (i.e., derelict house + empty house in good condition); 0 = sold house	0.333	0.477	0	1
Der	1 = derelict house (i.e., empty house in bad condition); 0 = o.w.	0.311	0.468	0	1
Independent variables					
Treat	1 = house located in treatment area; 0 = control area	1	0	1	1
EminentDomain	1 = house located in future eminent domain area; 0 = o.w.	1	0	1	1
Upzoning	1 = house located in future upzoning area; 0 = o.w.	0	0	0	0
DistStation	Distance to the closest station (metres)	301.0	154.0	25.7	625.0
LotSize	Lot size (square metres)	98.5	137.0	8.83	936.0
NarrowRoad	1 = frontal road width < 4m; 0 = o.w.	0.378	0.490	0	1
Neighbourhood$_n$	1 = house located in neighbourhood n (20 types); 0 = o.w.	–	–	0	1
		Observations = 45 (15 empty + 30 sold houses)			
Future Upzoning Area					
Dependent variables					
Emp	1 = empty house (i.e., derelict house + empty house in good condition); 0 = sold house	0.382	0.487	0	1
Der	1 = derelict house (i.e., empty house in bad condition); 0 = o.w.	0.309	0.464	0	1
Independent variables					
Treat	1 = house located in treatment area; 0 = control area	1	0	1	1
EminentDomain	1 = house located in future eminent domain area; 0 = o.w.	0	0	0	0
Upzoning	1 = house located in future upzoning area; 0 = o.w.	1	0	1	1
DistStation	Distance to the closest station (metres)	306.0	177.0	18.8	644.0
LotSize	Lot size (square metres)	81.4	53.4	9.12	390.0
NarrowRoad	1 = frontal road width < 4m; 0 = o.w.	0.487	0.501	0	1
Neighbourhood$_n$	1 = house located in neighbourhood n (20 types); 0 = o.w.	–	–	0	1
		Observations = 152 (58 empty + 94 sold houses)			
Control Area					
Dependent variables					
Emp	1 = empty house (i.e., derelict house + empty house in good condition); 0 = sold house	0.295	0.456	0	1
Der	1 = derelict house (i.e., empty house in bad condition); 0 = o.w.	0.218	0.413	0	1
Independent variables					
Treat	1 = house located in treatment area; 0 = control area	0	0	0	0
EminentDomain	1 = house located in future eminent domain area; 0 = o.w.	0	0	0	0
Upzoning	1 = house located in future upzoning area; 0 = o.w.	0	0	0	0
DistStation	Distance to the closest station (metres)	340.0	163.0	11.2	836.0
LotSize	Lot size (square metres)	94.7	105.0	2.86	1502.0
NarrowRoad	1 = frontal road width < 4m; 0 = o.w.	0.452	0.498	0	1
Neighbourhood$_n$	1 = house located in neighbourhood n (20 types); 0 = o.w.	–	–	0	1
		Observations = 1764 (521 empty + 1243 sold houses)			

Table A.3: Definitions of Variables and Descriptive Statistics for Rent Analysis

Variable	Definition	Mean	Std. dev.	Min	Max
Dependent variable of second stage					
Rent	Monthly advertised rent in Japanese yen	95939	42100	25000	1215000
Dependent variables of first stage					
Empty_{60}	Number of empty (often derelict) houses within 60m	0.734	1.28	0	13
Empty_{70}	Number of empty (often derelict) houses within 70m	0.971	1.55	0	16
Empty_{80}	Number of empty (often derelict) houses within 80m	1.31	1.99	0	18
Empty_{90}	Number of empty (often derelict) houses within 90m	1.64	2.41	0	23
Empty_{100}	Number of empty (often derelict) houses within 100m	1.99	2.84	0	28
Derelict_{80}	Number of derelict houses within 80m	1.01	1.65	0	14
Independent variables					
Houses_{60}	Number of houses within 60m	21.8	15.9	0	86
Houses_{70}	Number of houses within 70m	29.9	20.9	0	113
Houses_{80}	Number of houses within 80m	39.4	26.5	0	142
Houses_{90}	Number of houses within 90m	50.1	32.6	0	167
Houses_{100}	Number of houses within 100m	62.0	39.2	0	196
Age	Age of the building (years)	21.4	12.4	1	90
RoomArea	Floor space (square metres)	32.0	14.6	15	206
Station	Walking time to the closest station (minutes)	6.44	3.39	1	23
NeighComArea	Distance to the closest neighbourhood commercial area (metres)	42.1	69.3	0	444
DistFireSta	Distance to the closest fire station (metres)	608.0	254.0	31.2	1271.0
DistPoliceSta	Distance to the closest police station or *koban* (metres)	276.0	123.0	7.42	727.0
DistPostOffice	Distance to the closest post office (metres)	247.0	105.0	4.62	533.0
DistClinic	Distance to the closest clinic (metres)	81.0	52.4	1.75	335.0
DistSchool	Distance to the closest school (metres)	246.0	104.0	18.2	589.0
DistCareFacility	Distance to the closest care facility (metres)	160.0	79.1	8.43	443.0
Rooms	Number of rooms	1.27	0.534	1	8
Story	Floor level	3.83	3.23	-1	47
BuildingStory	Building story	6.55	5.01	1	54
FixedTermCon	1 = fixed-term contract; 0 = regular contract	0.00294	0.0542	0	1
Refurbished	1 = refurbished unit; 0 = o.w.	0.0293	0.169	0	1
ExistingRoad	1 = existing arterial road within 50m; 0 = o.w.	0.256	0.436	0	1
MostComRoad	1 = mostly completed arterial road within 50m; 0 = o.w.	0.0258	0.158	0	1
EmptyFlat	1 = an apartment building with empty units > 30% within 50m; 0 = o.w.	0.262	0.440	0	1
Structure_s	1 = building structure is s; 0 = o.w.	–	–	0	1
Year	1 = listed from 2011 to 2012; 0 = listed from 2016 to 2017	0.217	0.413	0	1
Neighbourhood_n	1 = unit located in neighbourhood n (20 types); 0 = o.w.	–	–	0	1
Excluded instrumental variables					
TreatArea_{60}	Treatment area coverage in the circle of radius 60m	0.0886	0.215	0	1.000
TreatArea_{70}	Treatment area coverage in the circle of radius 70m	0.0877	0.202	0	0.993
TreatArea_{80}	Treatment area coverage in the circle of radius 80m	0.0868	0.191	0	0.952
TreatArea_{90}	Treatment area coverage in the circle of radius 90m	0.0860	0.180	0	0.893
TreatArea_{100}	Treatment area coverage in the circle of radius 100m	0.0853	0.171	0	0.848
EminentArea_{80}	Eminent domain area coverage in the circle of radius 80m	0.0258	0.0653	0	0.533
UpzoningArea_{80}	Upzoning area coverage in the circle of radius 80m	0.0610	0.129	0	0.587
Dependent variable of placebo first-stage specification					
NewBuildings	Number of new buildings within 80m	7.02	4.79	0	32
TimeOnMarket	Time on market (days)	17.6	51.9	1	1547

Observations = 13936

B Rents' Response to Transport Improvement

We show that rents' response to transport improvement is observed only after the transport was placed in service, supporting exogeneity of our instrument, future arterial road development. Since no arterial road plans in Toshima were completed in the period for which we have sufficient rent data, we exploit another case of transport improvement in Tokyo including Toshima, which is the commencement of through running service of the Seibu Ikebukuro Line to Chinatown in Yokohama[B.1] in 2013. Through running service is a system where trains continue beyond the nominal terminal station of one railway line onto another line, often operated by different railway companies. This allows passengers to travel between two points without the need to transfer trains at the terminal station. In Japan, where railway networks are complex and stations can be extremely crowded during rush hours, the introduction of through running services can significantly improve passenger convenience and reduce travel time. The value of such improvements is particularly high in central Tokyo, where transferring at major stations often involves navigating through dense crowds and lengthy walking distances between platforms.

The Seibu Ikebukuro Line is one of the main rail lines that connects the centre and the suburbs of Tokyo with its terminal Ikebukuro Station in Toshima City. In Figure B.1, the Seibu Ikebukuro Line is depicted as an orange line and served its original through running service beyond Ikebukuro to Shibuya (technically, Shibuya ~ 01 Ikebukuro ~ 37 Kotake-Mukaihara ~ 06 Nerima ~ 26 Hanno) via Kotake-Mukaihara and Tokyo Metro Fukutoshin Line (brown line) since 2008. This means that, in reality, the Seibu Ikebukuro Line terminated at Shibuya. In 2013, it extended its through running service beyond Shibuya to Chinatown in Yokohama via Tokyu Toyoko Line (red and brown line) and Minatomirai Line (blue and brown line). Residents in areas along the Seibu Ikebukuro Line benefited from this extension as they no longer need to transfer at crowded Shibuya Station, for example, to commute to Yokohama or visit Chinatown for leisure.

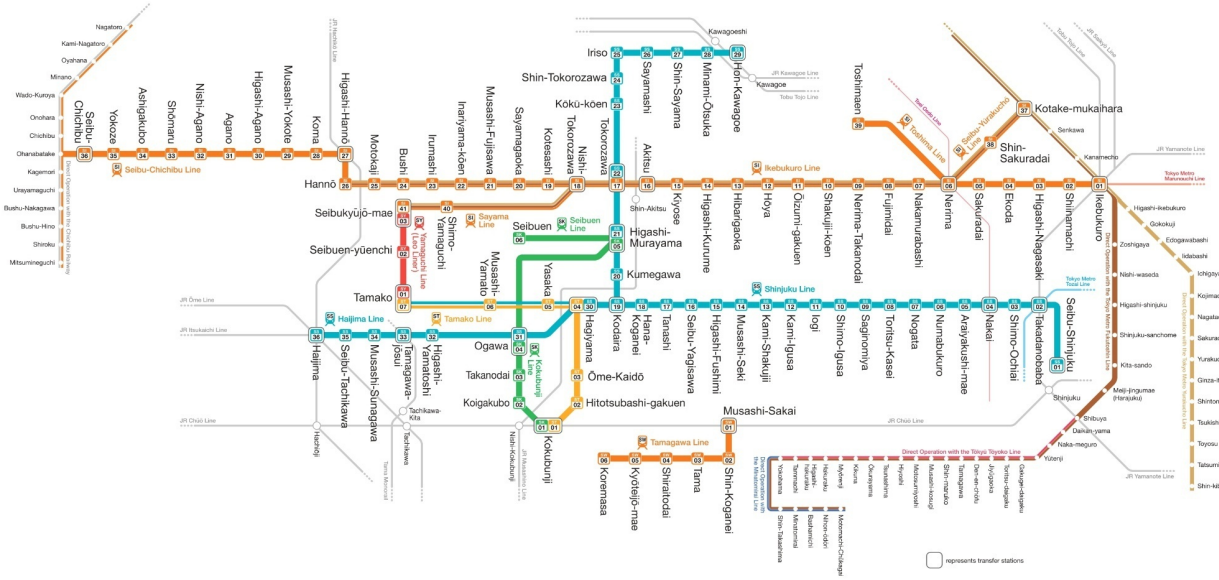
On the other hand, the Seibu Shinjuku Line, which is depicted as a light blue line in Figure B.1 (technically, 01 Seibu-Shinjuku ~ 29 Hon-Kawagoe and 36 Haijima), has no through running services and underwent no extensions beyond Seibu-Shinjuku, while it shares many similarities with the Seibu Ikebukuro Line. First, in the Japanese context, railway companies including Seibu Railway Co., Ltd. are privately owned and develop residential areas along their rail lines. This results in common characteristics of residential areas along the rail lines operated by the same company. Second, the Seibu Ikebukuro Line and the Seibu Shinjuku Line run mostly in parallel at a close distance (approximately 1 km to 5 km depending on sections) and both connect the centre and suburbs of Tokyo. Third, passengers on both lines can transfer at stations on Yamanote Line (loop line in central Tokyo connecting most of Tokyo's major stations). These attributes suggest that the areas along the Seibu Shinjuku Line are comparable to those along the Seibu Ikebukuro Line.

To evaluate the effect of the extension in 2013 on rents, we conduct an event study where we compare rental units whose nearest station is on the Seibu Ikebukuro Line (treatment group)[B.2] to those whose nearest station is on the Seibu Shinjuku Line (control group). Descriptive statistics are

[B.1] Chinatown in Yokohama is 37.2km away from the centre of Toshima (i.e., Ikebukuro).

[B.2] Because 02 Shiinamachi ~ 05 Sakuradai are not benefited from the extension, we exclude rental units whose nearest station is one of those stations from the sample.

Figure B.1: Seibu Railway Map



shown in Table B.1. Data cover rental units in the Special Wards in central Tokyo, including Toshima City, Bunkyo City, Itabashi City, Minato City, Nerima City, Nakano City, Shibuya City, Shinjuku City, and Suginami City, and other ordinary cities in suburbs of Tokyo, including Akiruno City, Akishima City, Higashi-Kurume City, Higashi-Murayama City, Higashi-Yamato City, Kiyose City, Kodaira City, Koganei City, Kokubunji City, Musashi-Murayama City, Musashino City, Nishi-Tokyo City, Fussa City, and Tachikawa City.

Let $Rent_{ft}$ be the monthly rent of unit f in year $t \in \{2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019\}$. $SeiIke_f$ is a dummy variable that takes the value of one if a rental unit f's nearest station is on the Seibu Ikebukuro Line. x_{ft} is a row vector of control variables representing unit f's attributes and neighbourhood characteristics. In addition, we control for unobservable location attractiveness using block-level fixed effects. Thus, our empirical model to evaluate the effect of the extension is as follows:

$$\ln Rent_{ft} = \alpha + \sum_{\substack{j=2011 \\ j\neq 2012}}^{2019} \beta_j \left(SeiIke_f \times \mathbf{1}\{t = j\} \right) + x_{ft}\gamma + \delta_b + \tau_t + u_{ft}, \tag{B.1}$$

where α and β_j are parameters, and γ is a column vector of parameters associated with a row vector of control variables x_{ft}. Block-level fixed effects are represented by δ_b, year fixed effects are represented by τ_t, and u_{ft} is an error term. We estimate equation (B.1) using the full sample, subsample of observations in central Tokyo, and subsample of observations in suburbs of Tokyo, respectively.

Table B.1: Definitions of Variables and Descriptive Statistics

Variable	Definition	Mean	Std. dev.	Min	Max
Treatment Group (Nearest Station is on the Seibu Ikebukuro Line)					
Dependent variable					
Rent	Monthly advertised rent in Japanese yen	94149	43285	12000	2010000
Independent variables					
Age	Age of the building (years)	20.7	12.1	1	99
RoomArea	Floor space (square metres)	38.2	17.9	15	454.0
Station	Walking time to the closest station (minutes)	7.13	5.61	1	601
Rooms	Number of rooms	1.55	0.769	1	27
Story	Floor level	3.31	2.51	-4	54
BuildingStory	Building story	5.55	3.91	1	55
Refurbished	1 = refurbished unit; 0 = o.w.	0.0152	0.122	0	1
Structure_s	1 = building structure is s; 0 = o.w.	–	–	0	1
τ_t	1 = listed in year t; 0 = o.w.	–	–	0	1
δ_b	1 = unit located in block b; 0 = o.w.	–	–	0	1
					Observations = 83085
Control Group (Nearest Station is on the Seibu Shinjuku Line)					
Dependent variable					
Rent	Monthly advertised rent in Japanese yen	84357	33810	10000	2230000
Independent variables					
Age	Age of the building (years)	22.0	11.6	1	99
RoomArea	Floor space (square metres)	37.2	17.6	15	436.1
Station	Walking time to the closest station (minutes)	7.19	4.55	1	201
Rooms	Number of rooms	1.59	0.780	1	21
Story	Floor level	2.75	1.68	-3	74
BuildingStory	Building story	4.42	2.26	1	69
Refurbished	1 = refurbished unit; 0 = o.w.	0.0125	0.111	0	1
Structure_s	1 = building structure is s; 0 = o.w.	–	–	0	1
τ_t	1 = listed in year t; 0 = o.w.	–	–	0	1
δ_b	1 = unit located in block b; 0 = o.w.	–	–	0	1
					Observations = 109947

Table B.2 and Figure B.2 present the estimation results. The estimations using the full sample and the subsample of observations in central Tokyo reveal that rents increased only after the through running service of the Seibu Ikebukuro Line extended in 2013 (i.e., expectations of this extension did not drive up rents in 2012 compared to those in 2011). The estimated effect of this extension was not statistically significant in suburbs of Tokyo, implying that residents in suburban areas, despite gaining direct access to Yokohama and Chinatown, may not frequently visit these destinations due to the considerable distance, thus receiving limited benefits from the extension.

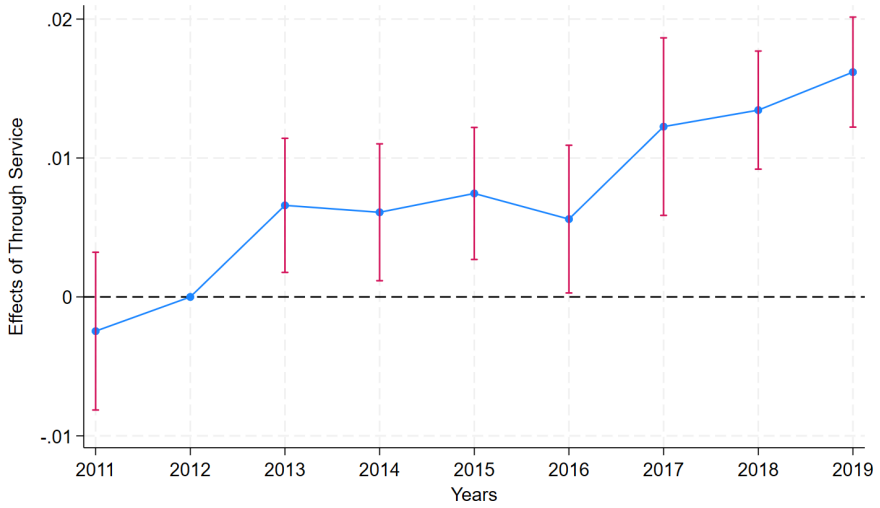
Table B.2: Estimation Results

	(1)	(2)	(3)
Dependent Variable: ln(Rent)			
$SeiIke \times \mathbf{1}\{t = 2011\}$	−0.00246	−0.00251	−0.0101
	(0.00290)	(0.00308)	(0.00859)
$SeiIke \times \mathbf{1}\{t = 2013\}$	0.00659***	0.00814***	0.000452
	(0.00246)	(0.00273)	(0.00537)
$SeiIke \times \mathbf{1}\{t = 2014\}$	0.00609**	0.00694**	−0.000535
	(0.00251)	(0.00282)	(0.00529)
$SeiIke \times \mathbf{1}\{t = 2015\}$	0.00745***	0.00662**	0.00888*
	(0.00242)	(0.00270)	(0.00516)
$SeiIke \times \mathbf{1}\{t = 2016\}$	0.00560**	0.00706**	−0.00150
	(0.00271)	(0.00302)	(0.00587)
$SeiIke \times \mathbf{1}\{t = 2017\}$	0.0123***	0.0116***	0.0101
	(0.00326)	(0.00359)	(0.00714)
$SeiIke \times \mathbf{1}\{t = 2018\}$	0.0134***	0.0126***	0.00794*
	(0.00217)	(0.00244)	(0.00461)
$SeiIke \times \mathbf{1}\{t = 2019\}$	0.0162***	0.0153***	0.00567
	(0.00202)	(0.00225)	(0.00434)
Sample	Full	Central Tokyo (Special Wards)	Suburbs of Tokyo (ordinary cities)
Control variables	Yes	Yes	Yes
Block FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	193032	144665	48367
Adjusted R^2	0.897	0.891	0.897

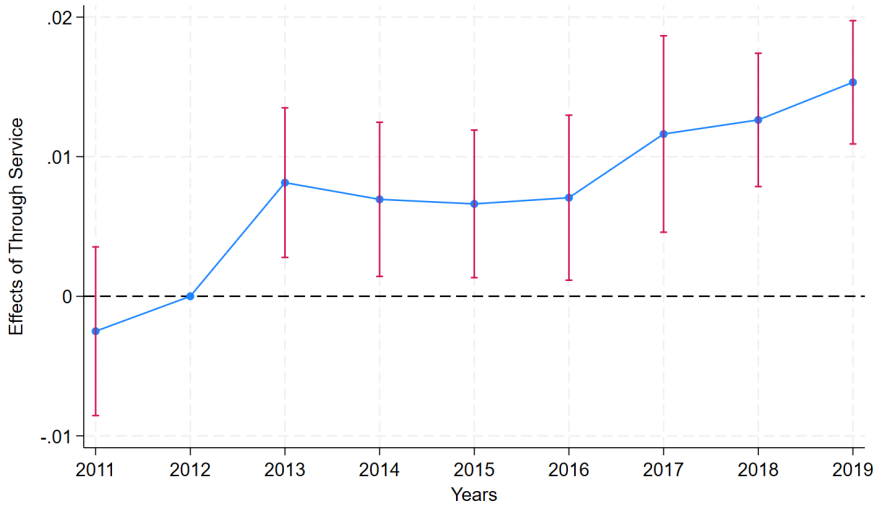
The unit of observation is the individual rental apartment unit. The dependent variable is the logarithmic value of the monthly rent of a rental apartment unit. ***, **, and * indicate 1 percent, 5 percent, and 10 percent significance levels, respectively. Heteroskedasticity-robust standard errors are in parentheses. Full results are available upon request.

Figure B.2: Estimation Results with 95% Confidence Intervals

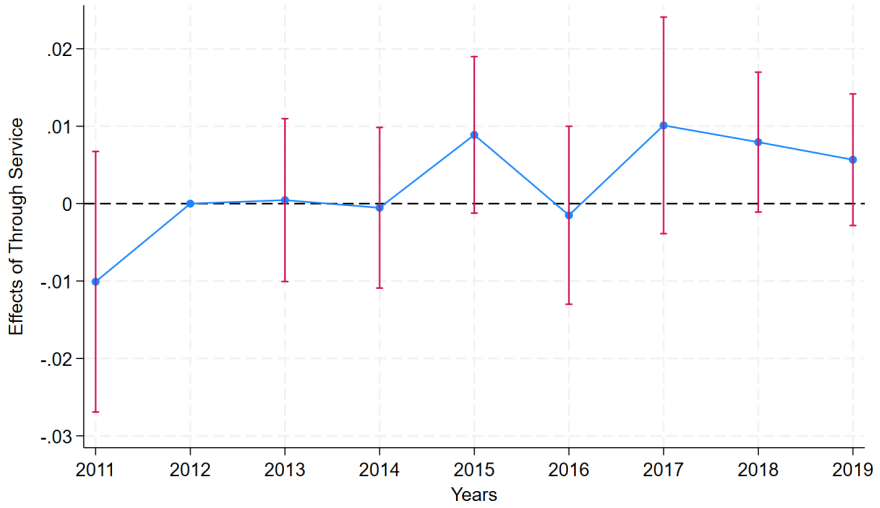
Panel A: Full sample



Panel B: Central Tokyo



Panel C: Suburbs of Tokyo



C Robustness Check for the Baseline 2SLS Results

We run a battery of robustness checks for our baseline 2SLS results in column (1) in Table 3, where we use the full sample and $r = 80$. Table C.1 presents the results of the robustness checks with variables defined in Table A.3.

Column (1) shows the results obtained replacing $Empty_{80}$ with $Derelict_{80}$. As defined in Table A.3, the former represents the number of empty houses, which are often derelict, within 80m, whereas the latter represents the number of derelict houses within 80m. Considering the discussion in Section 6.2, where we argue that the local average treatment effect estimated in Section 6.3 pertains specifically to the impact of derelict houses, using either variable should not largely affect the results. Although the estimated impact is slightly larger when using the latter variable, the confidence intervals largely overlap, showing that our baseline 2SLS results are robust to the choice of the variables.

Column (2) presents the results obtained after excluding rental units located in the treatment area (i.e., future eminent domain area + future upzoning area), addressing the potential concern that future road expectations may affect rents through not only the maintenance status of empty houses but also that of rental apartment units themselves. Results largely remain unchanged, confirming that our baseline 2SLS results are robust to this potential channel. Note that despite excluding rental apartment units in the treatment area, treatment status variation is preserved, as part of the sample lies within 80 metres of the treatment area's edge.

Column (3) presents the results obtained decomposing $TreatArea_{80}$ into $EminentArea_{80}$ and $UpzoningArea_{80}$. As these two variables are highly correlated ($\rho > 0.9$), presumably due to multicollinearity, no statistically significant relationship between the future upzoning area coverage and the number of empty houses is observed. Having said that, the future eminent domain area coverage is still statistically significant, and the first-stage estimation is strong enough. The second-stage result largely remains unchanged, ensuring that our baseline 2SLS results are robust to the definition of the instrumental variable(s).

Table C.1: Robustness Check of the Baseline 2SLS Results (Rent Analysis)

	(1) 2SLS	(2) 2SLS	(3) 2SLS
Dependent Variable: ln(Rent)			
Second stage			
Empty_{80}	–	**-0.0186****	**-0.0259*****
	–	**(0.0090)**	**(0.0085)**
Derelict_{80}	**-0.0187****	–	–
	(0.0095)	–	–
First stage			
TreatArea_{80}	0.1205***	0.2466***	–
	(0.0182)	(0.0399)	–
EminentArea_{80}	–	–	0.1286***
	–	–	(0.0390)
UpzoningArea_{80}	–	–	0.0102
	–	–	(0.0440)
Sample	Full	Without observations in the treatment area	Full
Control variables	Yes	Yes	Yes
Neighbourhood FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	13936	12691	13936
KP F-statistic	43.9	38.2	28.1

The unit of observation is the individual rental apartment unit. The dependent variable of the second stage is the logarithmic value of the monthly rent of a rental apartment unit. ***, **, and * indicate 1 percent, 5 percent, and 10 percent significance levels, respectively. Heteroskedasticity-robust standard errors are in parentheses. **Bold** indicates that Empty_{80} and Derelict_{80} are instrumented. Control variables include Houses_{80}. Full results are available upon request.

D Distance-Decaying Disamenity Effects

Tables D.1 and D.2 respectively show OLS and 2SLS results of rent analysis with various r using the full sample. Same as in our baseline results in Table 3 column (1), OLS estimates tend to be biased upward, implying that derelict houses are more likely to appear in high-priced treatment areas. The first-stage estimations of 2SLS show our instruments positively correlate with empty house locations and are strong enough regardless of the value of r. The second-stage estimations of 2SLS results show that the estimated effects of one additional derelict house on rents are statistically and economically significant, aligning with our findings from the baseline results.

Moreover, the second-stage results show distance-decaying disamenity effects, complementing our baseline results. Specifically, the results in Table D.2 show that the estimated effects diminish as r approaches 100. These results align with intuition that potential renters would care less about derelict houses 100m away than those 60m away.

Table D.1: Results of Rent Analysis (Full Sample: OLS)

	(1) OLS	(2) OLS	(3) OLS	(4) OLS	(5) OLS
Dependent Variable: ln(Rent)	$r = 60$	$r = 70$	$r = 80$	$r = 90$	$r = 100$
Second stage					
Empty_r	-0.0033***	-0.0037***	-0.0032***	-0.0032***	-0.0029***
	(0.0008)	(0.0007)	(0.0006)	(0.0005)	(0.0004)
Control Variables	Yes	Yes	Yes	Yes	Yes
Neighbourhood FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	13936	13936	13936	13936	13936
Adjusted R^2	0.891	0.892	0.892	0.892	0.892

The unit of observation is the individual rental apartment unit. The dependent variable of the second stage is the logarithmic value of the monthly rent of a rental apartment unit. ***, **, and * indicate 1 percent, 5 percent, and 10 percent significance levels, respectively. Heteroskedasticity-robust standard errors are in parentheses. Control variables include Houses_r. Full results are available upon request.

Table D.2: Results of Rent Analysis (Full Sample: 2SLS)

Dependent Variable: ln(Rent)	(1) 2SLS $r = 60$	(2) 2SLS $r = 70$	(3) 2SLS $r = 80$	(4) 2SLS $r = 90$	(5) 2SLS $r = 100$
Second stage					
Empty_r	**-0.0248****	**-0.0245****	**-0.0163****	**-0.0146***	**-0.0129***
	(0.0116)	**(0.0124)**	**(0.0083)**	**(0.0077)**	**(0.0069)**
First stage					
TreatArea_r	0.0971***	0.0936***	0.1381***	0.1520***	0.1699***
	(0.0157)	(0.0177)	(0.0201)	(0.0232)	(0.0265)
Control variables	Yes	Yes	Yes	Yes	Yes
Neighbourhood FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	13936	13936	13936	13936	13936
KP F-statistic	38.3	27.9	47.4	42.8	41.2

The unit of observation is the individual rental apartment unit. The dependent variable of the second stage is the logarithmic value of the monthly rent of a rental apartment unit. ***, **, and * indicate 1 percent, 5 percent, and 10 percent significance levels, respectively. Heteroskedasticity-robust standard errors are in parentheses. **Bold** indicates that Empty_r is instrumented. Control variables include Houses_r. Full results are available upon request.

E Areas Distant from Placebo Facilities

We conduct another subsample analysis using observations distant from placebo facilities (i.e., clinics and care facilities), which are not expected to alleviate the disamenity effects of derelict houses. This analysis aims to provide supportive evidence that the pronounced effects estimated in Section 6.3, which we attribute to low accessibility to the public safety services, are not contaminated by other confounding factors, such as peripheral location within neighbourhoods represented by the distance to public facilities in general.

Table E.1 shows OLS and 2SLS results of rent analysis using a subsample consisting of observations distant from clinics and a subsample consisting of observations distant from care facilities. In areas with low accessibility to these placebo public services, we expect that potential renters' concerns about risks of fire and crime attributable to nearby derelict houses would not necessarily be pronounced.

First-stage results indicate that our instrumental variables positively correlate with empty house locations and are strong enough, whereas the second-stage results indicate no statistically significant effects of the proximity to derelict houses on rents. We can interpret these results to mean that the disamenity effects of derelict houses on nearby rents are not pronounced in areas distant from the placebo facilities. These results align with our expectation and contrast with the statistically and economically significant effects in areas with low accessibility to the public safety services.

Table E.1: Effects of Derelict Houses on Nearby Rents in Areas Distant from Placebo Facilities

	(1)		(2)	
	OLS	2SLS	OLS	2SLS
Dependent Variable	ln(Rent)	ln(Rent)	ln(Rent)	ln(Rent)
Second stage				
Empty_{80}	-0.0018**	**0.0076**	-0.0026***	**-0.0148**
	(0.0008)	**(0.0084)**	(0.0009)	**(0.0134)**
First stage				
TreatArea_{80}	–	0.2570***	–	0.1253***
	–	(0.0304)	–	(0.0242)
Sample	Subsample (Distant from Clinics)		Subsample (Distant from Care Facilities)	
Control Variables	Yes	Yes	Yes	Yes
Neighbourhood FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Adjusted R^2	0.878	–	0.887	–
KP F-statistic	–	71.3	–	26.8
Observations	6106	6106	6722	6722

The unit of observation is the individual rental apartment unit. The dependent variable of the second stage is the logarithmic value of the monthly rent of a rental apartment unit. ***, **, and * indicate 1 percent, 5 percent, and 10 percent significance levels, respectively. Heteroskedasticity-robust standard errors are in parentheses. **Bold** indicates that Empty_{80} is instrumented. Full results are available upon request.

F Placebo First-Stage Specification

There are two potential channels that would threaten the exogeneity of our instrument. First, one may argue that the future upzoning expectations induce incremental urban transitions in the surrounding areas and thus positively affect rents through urban renewal even before the road construction. Second, the future expectations — though not affecting potential renters' willingness to pay — may affect their arrival rate or landlords' search behaviour. If this is the case, the future expectations would indirectly affect rents via vacancy duration of rental units (i.e., time-on-market).

To address these concerns, we run placebo first-stage specifications, in which we replace the number of empty houses with the number of nearby new buildings and each rental unit's time-on-market, respectively. Table F.1 presents the placebo first-stage results. Columns (1) and (2) show that neither nearby redevelopment nor rental unit's time-on-market has statistically significant relationship with the future road construction, meaning that we have no evidence of the abovementioned two exclusion channels.

Table F.1: Placebo First-Stage Specification

	(1) OLS	(2) OLS
Dependent Variable	NewBuilding_{80}	TimeOnMarket
TreatArea_{80}	-0.0360	0.1556
	(0.0335)	(0.4722)
Control Variables	Yes	Yes
Neighbourhood FE	Yes	Yes
Year FE	Yes	Yes
Adjusted R^2	0.486	0.019
Observations	13936	13936

The unit of observation is the individual rental apartment unit. The dependent variables are the time on market of each rental apartment unit and the number of new buildings within 80m, respectively. ***, **, and * indicate 1 percent, 5 percent, and 10 percent significance levels, respectively. No statistically significant relationships were found (i.e., the exclusion restriction is not violated at least through the nearby redevelopment and the time-on-market pathways). Heteroskedasticity-robust standard errors are in parentheses. Full results are available upon request.

G Extended Literature Review

Causes of Housing Dereliction

The mechanism behind the occurrence of empty houses has been examined in economics literature, where most existing studies focus on the context of urban decline. For instance, the literature has established that housing vacancy rates rise under housing price depreciation due to market frictions that prevent prices from falling sufficiently to clear the market (Molloy, 2016; Glaeser and Gyourko, 2005; Goodman and Ittner, 1992; Genesove and Mayer, 2001; Anenberg, 2016; Guren, 2018). Under the same circumstances, loss aversion can also explain the occurrence of empty houses (Fritzsche and Vandrei, 2018). Housing abandonment and dereliction caused by industrial decline and tax delinquency in blighted areas has also been examined (Arsen, 1992; O'Flaherty, 1990; Scafidi et al., 1998; White, 1986).

In contrast, empty houses that coexist with high housing demand have been explained in different contexts. One potential explanation is given by the search and matching process. Previous studies show that mismatch caused by restrictive planning systems (Cheshire, Hilber and Koster, 2018), imperfect information (Shimizu, Nishimura and Asami, 2004), and housing market cyclicality (Díaz and Jerez, 2013; Head, Lloyd-Ellis and Sun, 2014; Ngai and Tenreyro, 2014) can cause housing vacancy. Another explanation would be a myopic development decision in the past causing today's housing glut in areas where strong housing demand growth was once anticipated (Zheng, Zhang, Yin and Wu, 2025). However, these frameworks do not necessarily provide a reasonable explanation for inactive vacancy (i.e., withdrawal from the housing market) in currently high-demand areas.

Owners' strategic behaviour stemming from their inherent preferences helps understand the mechanism of inactive housing vacancy and dereliction in a prosperous area. For example, combining owners' idiosyncratic participation constraints with the search and matching process well explains the inactive housing stock despite a high housing demand (Segú, 2020). This relates to another study showing that owners' high reservation prices may cause derelict houses in a prosperous area (Kanayama and Sadayuki, 2021). Owners may also prefer to wait until matching with a tenant with high willingness to pay (Moszkowski and Stackman, 2023).

Literature on real options also help explain prolonged property vacancy under a demand growth. The foundational work in investment decisions shows that, under uncertainty, traditional NPV rules are insufficient and investment should be delayed until benefits substantially exceed costs rather than merely when NPV becomes positive (McDonald and Siegel, 1986; Ingersoll and Ross, 1992). Its applications to vacant land development has yielded consistent theoretical and empirical evidence: increased price volatility (i.e., uncertainty) delays development timing (Capozza and Li, 1994; Cunningham, 2006; Bulan et al., 2009). Later, the literature has been extended beyond simple development decisions to include multiple development options, where introducing alternative uses can reduce irreversibility and accelerate initial investment (Lin, 2024), and to building replacement decisions where structural features affect property prices depending on teardown probability (McMillen and O'Sullivan, 2013).

Moreover, the real options literature shows that development delays occur not only under uncertainty but also under certainty when cash flows are growing and capital intensity is variable (Capozza

and Li, 2002), with counterintuitive effects where high growth rates can suppress investment (Lange and Teulings, 2024). These findings complement seminal urban economics research showing that, under perfect foresight, the ability to optimise timing and residential density in response to anticipated income growth leads to delayed development (Wheaton, 1982).

Disamenity Effects

The disamenity effects of derelict houses on nearby housing values have been examined in both blighted and prosperous areas. To this end, researchers normally use hedonic price functions and demonstrate significant price declines associated with nearby derelict houses (Sadayuki et al., 2020; Suzuki et al., 2022; Paredes and Skidmore, 2017; Whitaker and Fitzpatrick, 2013). However, the absence of compelling instruments and insufficient data have compelled the researchers to conduct cross-sectional analysis despite endogenous derelict house locations and thus a causal relationship has not been fully disentangled.

In the Japanese context, previous studies show that proximity to an empty house, which is often derelict, is negatively associated with housing rents and prices. More specifically, in central Tokyo, rents of a housing unit that is located within 50m from an empty house is approximately 1.7% lower than that of a housing unit with no nearby empty houses (Sadayuki et al., 2020). Similarly, in a depopulating city on the periphery of the Tokyo metropolitan area, one additional long-term empty house within 50m is associated with a 3% decline in property transaction prices (Suzuki et al., 2022).

Aligning consequences are also seen in the US context. In Detroit, an additional empty house and dilapidated house within 0.05 miles (80.5m) is associated with a 5% and an 8% decline, respectively, in property transaction prices (Paredes and Skidmore, 2017). Similarly, an additional empty property within 500 ft (152.4m) reduces housing transaction prices by 1% to 2% in Cuyahoga County, Ohio (Whitaker and Fitzpatrick, 2013).

The foreclosure literature, where causal inference on the impact of foreclosed properties on nearby property values has been commonly conducted, is also relevant. While some pieces of the literature show the disamenity effects of foreclosures (Harding et al., 2009; Campbell et al., 2011; Gerardi et al., 2015; Bak and Hewings, 2017), others suggest that supply-side shocks through REO sales drive price declines (Anenberg and Kung, 2014; Hartley, 2014). For instance, Hartley (2014) demonstrates the dominance of this supply-side effect under the assumption that the markets of houses and apartment units are separated. He assumes that the effect of a single-family foreclosure on nearby multi-family housing prices reflects a disamenity effect, as potential buyers in the apartment market do not regard a single-family foreclosure as a substitute. He then shows the absence of disamenity effects in Chicago by using a difference-in-difference approach.

Uniqueness of This Study

Regarding the paradoxical occurrence of housing dereliction in prosperous areas, the role of prosperity-driven urban policies that may unintentionally induce owners' decisions to neglect their houses has yet to be explored. For instance, future urban regeneration may incentivise housing owners to delay

redeveloping their properties. We theoretically and empirically examine this aspect, contributing to understanding the property neglect paradox despite prosperity. Our approach sheds light on strategic behaviour of owners facing dynamic discrete choices, which is somewhat similar but technically different from those used in the real options literature. This approach allows us to obtain testable predictions and validate them at a granular level by exploiting fine-grained within-city variation, which standard option value studies does not necessarily address.

Subsequently, given the lack of causal inference about the disamenity effects of derelict houses, we quantify them leveraging future urban regeneration as the instrument supported by our analysis on the causes of housing dereliction. Although the derelict house data we use are unique and granular, the data are still insufficient to conduct a panel data analysis and compel us to conduct cross-sectional analysis as in the previous literature. Thus, this instrument is crucial to estimate the disamenity effects addressing the potentially endogenous derelict house locations.